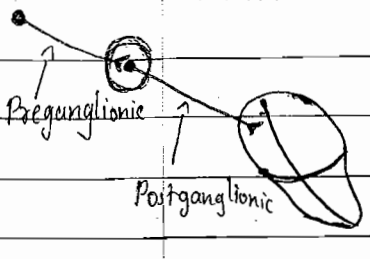
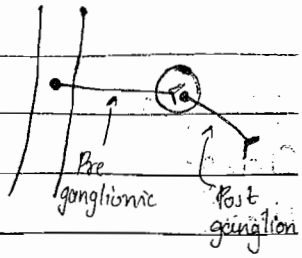
ANS

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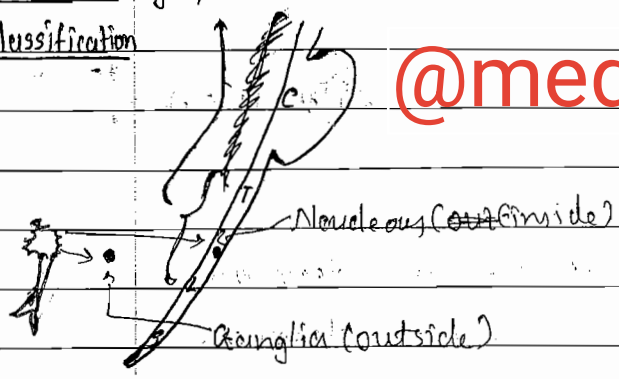
Autonomic Nerveous System: [ANS]



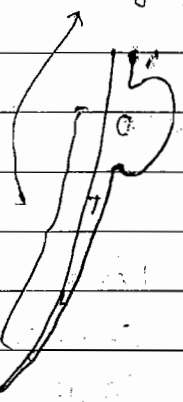


ANS

Anatomical classification: Sympathetic



Parasympathetic



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	Sympathetic	Parasympathetic
Nerve Situation	Thoraco-Lumbar nerves.	- Cranio-Sacral Nerves.
Ganglia Location	- Near the spinal cord (away from the organ)	- Near the organ
Preganglionic fibres	Short	Long
Postganglionic fibres	Long	Short
Neuro Transmitters	Pre - Ach (Secretion) Post - Ach	Pre - Ach Post - Adrenaline / Noradrenaline)
Receptors	Adrenergic	Muscarinic and Nicotinic

Adrenergic: α, β

- α → α_1 (Postsynaptic), α_2 (Presynaptic)
- β → β_1 (Heart), β_2 (GIT) (Vessels), β_3 (adipose tissue)

Muscarinic and Nicotinic

- M_1 (GIT), M_2 (Heart), M_3 (Vessels)
- Nicotinic → Nmus., Nnico.

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Physiological Changes	Parasympathetic	Sympathetic
Heart	Heart rate ~~Increase~~ Reduce	Heart Rate Increase
B.P.	B.P. decrease	B.P. increase
Respiratory System	Broncho constriction	Bronchodilatation
GIT	Peristalsis increase (gastric motility) Secretion increase (Saliva, Gastric Juice)	Peristalsis decrease Secretion decrease
Bladder	Increase Urination	Decrease Urination
Eye	Miosis (Pupil Contraction)	Mydriasis (dilatation)
CNS	depressive	Stimulatory

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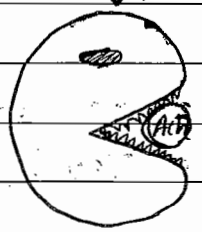
✯ Cholinergic System:-

-> Cholinergic agonist ← Parasympathomimetics
Cholinergic (Same as Parasympathetic)

Anti cholinergic → Parasympatholytic
Parasympathetic → agonist

-> Classification:-

Cholinoesters	Alkaloids	Anti-Cholinesterase (inhibite)
- Acetyl choline	- Pilocarpine	Reversible / Irreversible
- Bethanechole		- Neostigmine
- Methacholine		- Physostigmine
- Carbechole		- Pyridostigmine



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☆ Classification of Cholinergic Drug :-

- **Cholinergic Drug**
  - **Directly acting drugs**
    - **Cholinesters**
      - Acetylcholine
      - Bethanachol
      - Carbachol
      - Methacholine
    - **Alkaloids**
      - Pilocarpine
      - Muscarine
  - **Indirectly acting drugs**
    - **Anticholinesterase**
      - **Reversible**
        - Neostigmine
        - Physostigmine
        - Pyridostigmine
        - Rivastigmine
        - Ednophonium
      - **Irreversible**
        - Malathione
        - Parathione
        - Diazinone
        - (Tubun, Sarin, Somen) nerve gas.

☆ Pharmacological Base of action of the cholinergic drug ? (Pilocarpine, Neostigmine).

↝ 1.] CVS :-

→ Cholinergic (Ach) drug binds on M_2 receptors of heart.

→ K^+ channels open ⟶ hyperpolarisation

↓

SA, AV nodes secretion decrease

↓

Heart Rate decrease ⟵ Force of contraction decretion

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2.] Respiratory System:-

→ Cholinergic drug binds on M_3 receptors of smooth muscle of Trachio-Bronchal tree.
→ Smoot muscle contraction occurs.
→ Lumen become short.
→ That's why it is contraindicated (unused) in asthma.

3.] GIT:-

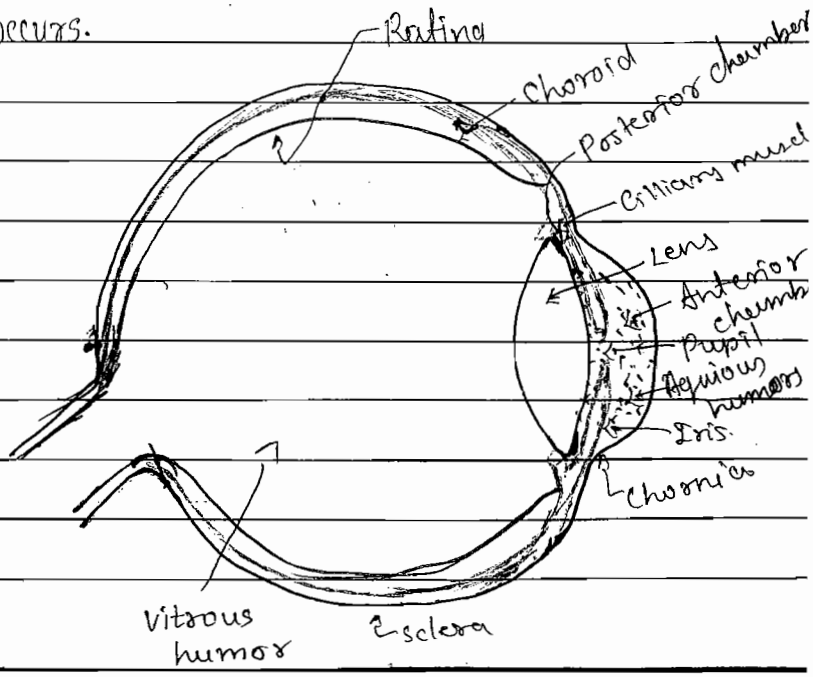
→ Cholinergic drug binds on M_1 receptors of GI tract.
→ It stimulate contraction of smooth muscle of GIT.
→ Increased peristalsis movement.
→ Stimulate glands - Secretions are increased.

4.] Bladder:-

→ Cholinergic drugs binds on M_3 of bladder smooth muscles (detrosour muscle).
→ Contraction of deterosour muscle and relaxation of bladder trigone.
→ and urination occurs.

5.] Eye:-

Aquious humours production increase
or
drainage decrease



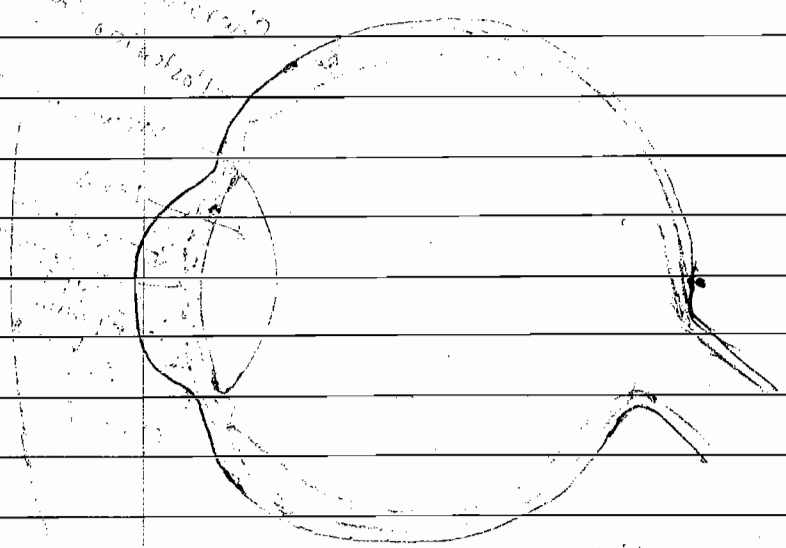
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→ Cholinergic drugs binds on iris and ciliary smooth muscles receptors.
and increase contraction of iris smooth muscles.
→ Pupil → Contricted (diameter reduce) (Miosis)
→ Trabecullar outflow of aquous humor increase → decrease intra occular pressure.
→ It is used in glaucoma.

(Ques: Pilocarpine used in Glaucoma ?)

★ Pharmacological base of action of the indirect cholinergic drug?

→ Anticholinesterase
→ They bind with cholinergic enzyme.
→ Inhibite the enzyme.
→ Ach destruction reduce.
→ Ach concentration increase.
→ Cholinergic action.



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☆ Uses of Cholinergic Drugs? (~~Anticholinergics~~):-

(Anticholinesterase Drug)

[Niostigmine;
Physostigmine;
Pyridostigmine;
Edrostigmine;
Rivastigmine]

→ 1.] Myesthenia Gravis:-

- It is an autoimmune disorder.
- Antibody is formed against acetylcholine Receptors.
- Muscle Paralysis (Muscle Weakness).
- When there is involvement of Respiratory muscles.

↓

Myesthenia Crysis.

aus Treatment:-

- Anticholinesterase drugs (Neostigmine, Pyritostigmine).

• Reasons:-

1.] Anticholinesterase inhibite Cholinesterase enzyme
  • So, Acetylcholine concentration increase.
  • Symptoms of mystemia gravis Resolved.

ii.] Corticosteroids → Reduce antibody production.

iii.] Plasmapharesis → Removes the antibody from blood.

iv.] Thymectomy → Removal of thymus gland.

aus 2.] Glaucoma:-

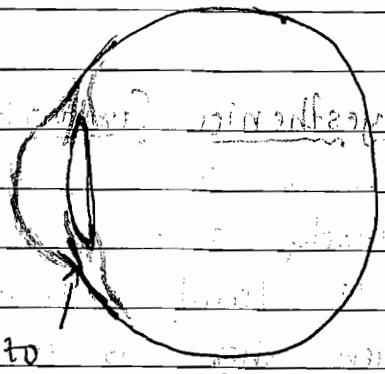
→ Group of disorders in which there is damage to the optic nerve and retina.

→ Most Commonly due to increase intra occular tension.

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→ Aquous humour production increase.
→ Outflow decrease.
→ Accumulation of fluid in chamber.
→ Increase pressure on lens.
→ Pressure transfered back and damage to retina and optic nerve.
→ Vision problem.



∴ Types :-

iritocorneal angle open — iritocorneal angle close

→ Pilocarpine is used.
→ Directly acting.
→ Binds on M_3 receptors of Iris (Smooth Muscle)
→ Increase contraction of smooth muscle.
→ Miosis.
→ Increase outflow of Aquous humour.
→ Reduce intra occular tenson.
- So, used in Glaucoma.

ans 3.] Post operative Urinary Retention :-
→ Neostigmine can be used for postperative
→ Urinary Retension.
→ Beacause → It increase bladder contraction.
→ So, Urine is passed.

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ans 4.] Post operative Reversal of Neuromuscular Blocker

→ If the muscles do not revert to contraction stage after operation.

→ Neostigmine can be given.

→ It will revert the neuromuscular block by increasing acetylcholine Concentration of neuromuscular junction.

Neostigmine	Physostigmine
→ Water Soluble	→ Lipid Soluble
→ Can not cross Blood Brain Barrier.	→ Can cross Blood Brain Barrier.
→ So, Only Peripheral effect.	→ So, Central & Peripheral effect.

ans 5.] Belladona Poisioning :- (Exam)

→ Physostigmine is used.

→ It is having central and peripheral effect.

ans 6.] Alzheimer's Diesease :-

→ In which Acetyl Choline (Ach) concentration is Reduced in Brain.

· Neuronal degredation.

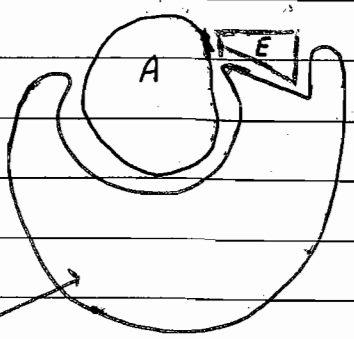
→ Dementia (Memmory Loss).

→ Donepezil, Rivastigmine can be used for this.

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☆ Organophosphate Poisioning ?



→ Anticholinesterase Poisioning.

→ Agents:-
- Organophosphate which binds irriversibly with Cholinesterase enzyme

- Dyflos — Gas form - Tabun
- Parethione - Sarin
- Malathione - Somen

→ Exposure:-
- Inhalation
- Injection
- Skin contect

→ Clinical Symptoms:-

↝ Bredicardia (Decrease Heart Pressure).
- Blood Pressure fall down
- Severe bronchoconstriction (Difficulty in Respiration)

↝ Abdominal Cramps (increase peristalsis).
- Urine pass spontenandy
- Stool Pass
- Salivation increase.

↝ Miosis (Pin Point Pupil)

↝ Drowsiness
- Coma → Death (Due to Respiratory paralysis).

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Diagnosis :-

- History of exposure.
- Clinicle features
- Cholinesterase Concentration test.

Treatment :-

- Removal of person from further exposure.
- Remove the cloths, Wash Patient with soap and water.
- Airway maintainance ; Mechanical breathing
  - O_2 therapy
  - ~~Trachea~~ Intubation (Tube insertion in R.T.).
    Respiratory Tract.
- Circulation - $\overrightarrow{IV}$ fluid
  - $\overrightarrow{IV}$ diazepam - Calm and prevention of convultion.
- Gastric Lavage.

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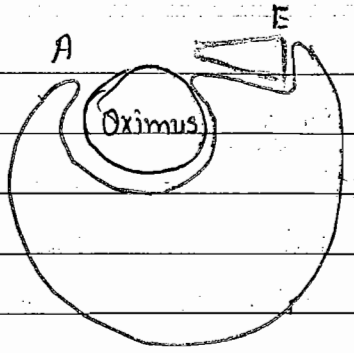
→ Specific Treatment :-

1.1 Anticholinergic :-

- Atropin → 2 mg $\overrightarrow{IV}$ → Repeat at 10 to 15 minutes interval.
- In 24 hrs → 200 mg maximum.
- Use → upto 7 to 10 days

Check For atropinisation → • Dilated Pupil
• Tachycardia.

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→ 2.] Oximus:-

- Oximus binds with cholinesterase enzyme at Anionic site.
- Dephosphorylate the enzymes [Removes the ~~organoorganophosphate~~ organophosphate compound].
- activate enzyme.
- It should be given within 24 hours of organophosphate poisioning.
- So, reactivation is not possible.
- Pralidoxime Dose :- 2 gms - at 8 hrs internal
  - upto 7 days.
  - Dose will be dependant on the patient clinicle status.

★ Classification of anticolinergic Drugs?

→ Natural Anticholinergic Drugs.
- Atropin
- Hyoscine.

→ Semisynthetic Anticholinergic Drugs.
- Homotrophine
- Ipratropium Bromide
- Triotropium Bromide

→ Synthetic anticholinergic Drugs.

a.] Antisecretory and antispasmodic.
- Dicylomine
- Glycopyrolate

b.] Mydriatic.
- Cyclopentolate
- Tropicamide

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→ c.] As Antiparkinsonism Drug:-
- Benzexol
- Biperidine
- Procyclidine

→ d.] Vesicoselective:-
- Oxybutinine.

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★ Atropine → Pharmacological Base of Action:-

↳ 1.] CVS:-
- Increase the force of contraction } → because of M_2 receptor blockage.
- Increase the heart rate.
- AV node conduction fast.

↳ 2.] CNS:-
- excitement
- Alterness
- Inhibite cholinergic action of basal ganglia.

Cholinergic ⟷ Dopaminergic

↳ 3.] Respiratory:-
- Smooth muscle of trachio-bronchial tree → Relaxation → Bronchodilatation
  → Because of M_3 receptor blockage.

↳ Glands:-
- Reduce secretion of glands. [Salivary Glands, GIT Glands]

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5.] GIT :-
- Relaxation of GIT smooth muscles. } → Because of M_3 receptor blockage.
- Reduce peristalsis.
- constipation.

6.] Excretory :-
- Urine Retention } → Because of M_3 receptor blockage.
- Bladder Relaxation

7.] Eye :-
- Block M_3 receptor of constrictor pupil
- Mydriasis (dilatation of pupil).
- Passive Dilatation.
- lens become flate.
- Vision is fixed for far vision.
- Acomodation reflex → Lost.

★ Que :- Atropin :- Substitute ?

→ - Selective drugs are used for perticular action on an orgen.
- It reduce adverse reaction. action become more selective.

Eye :
Cyclopentolate and tropicamide → Opthalmic examination
- Fundoscopy
- Retinoscopy.

- Atropin having → Longer duration
→ blurring of vision
→ Accomodation Reflex lost } → longer duration.

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- Cyclopentolate → 24 hr.
- Tropicamide 6-8 hrs. duration → Preferred of action.

~ Antispasmodic :-

- **Dicyclomine**
  - More selective for GIT smooth muscle.
  - So, used as antispasmodic (Relaxation of smooth muscle).

- **Dysmanorrhea**
  - Dicyclomine is used.

~ Precinesthetic medication :-

- **Glycopyrrolate**
  - Having more selective action to block secretion.

~ Parkinson's Diesease :-

- **Benxexol**
- **Biperidine**
- **Procyclidine**

→ Can work on CNS
more selective

- So, used in a parkinson's diesease to reduce cholinergic activity of basal ganglia.

~ Asthma :-

- **Ipratropium Bromide**
- **Tiotropium Bromide**
  - Bronchoselective action.
  - Used in as inhalation.

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~ Oxybutaminel.
- Vasoselective
- More on urinary bladder
- Used for bladder spasm.

↳ Receptor M_3 is in the bladder

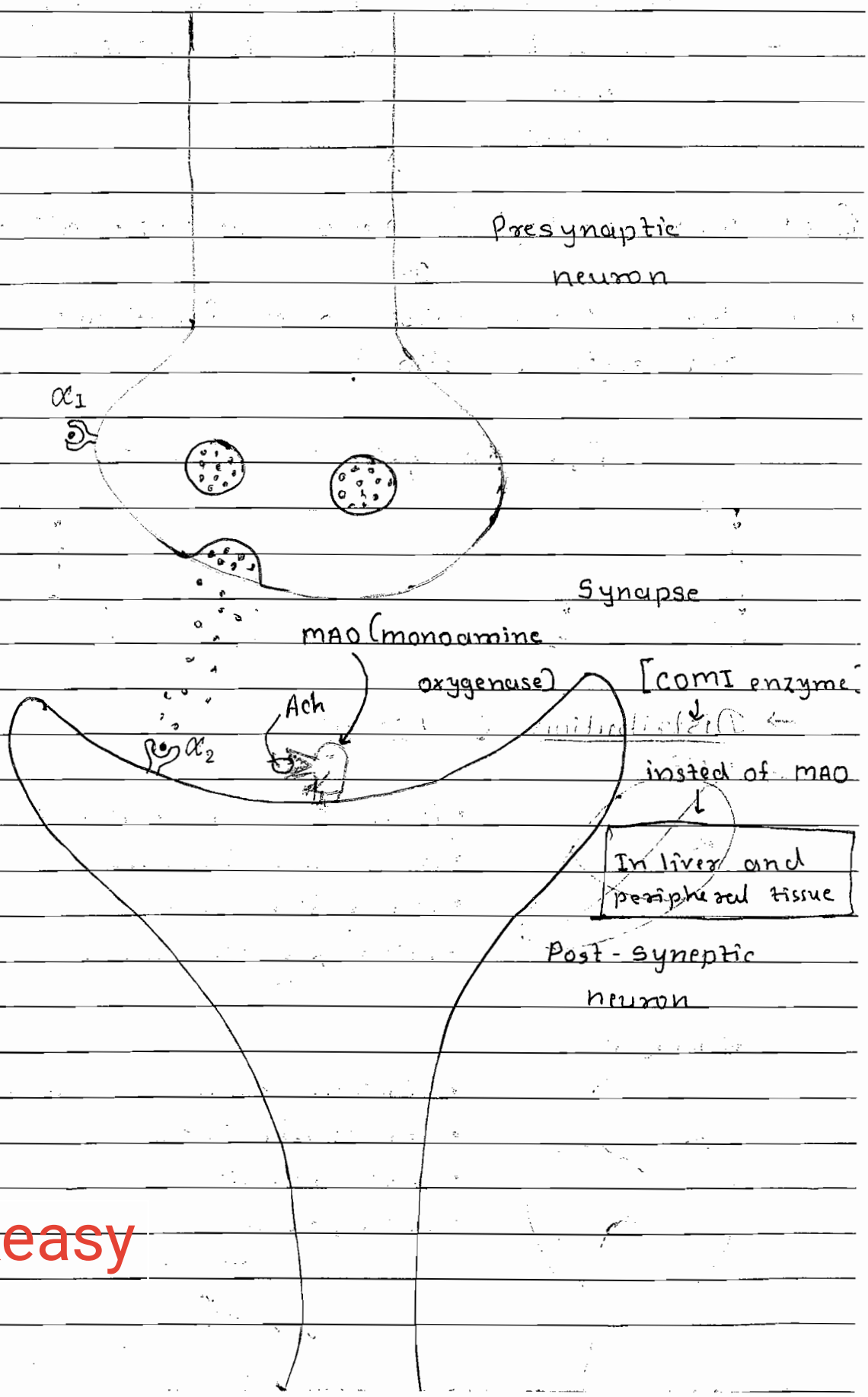
~ Vagolytic:
- Atropin - Vagus nerve action → block
  AV node conduction fast.
- Cardiac Block → It is used.
- Bradycardia.

Que → Adverse Drug Reaction of Anti-cholinergic Drug:
Vsq
1 marks.
- Dryness of mouth
- Constipation.
- Tachycardia.
- Cerebiac arrythimia
- Excitation
- Hulutination.
- Urinary Retention, difficulty of micturation.
- Accomodation Reflex - lost
  Blurring of vision.

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☆ Adrenergic transmission and metabolism?



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→ Drugs directly acting on adrenergic receptors known as directly acting adrenergic drug.

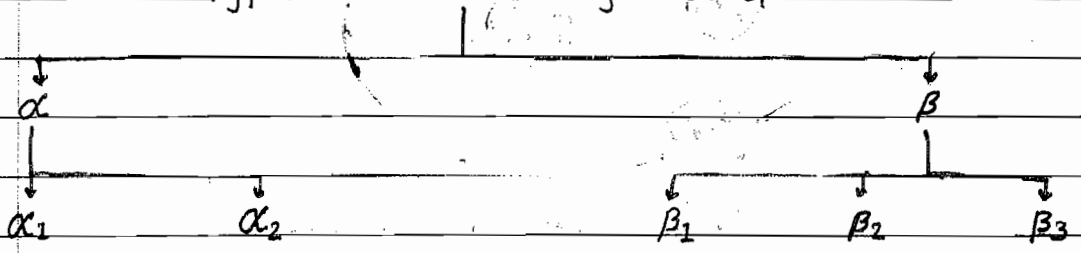
→ Metabolism of adrenergic agonist
- MAO
- COMT

Que 1- Pharmacological action of adrenaline ?

Or

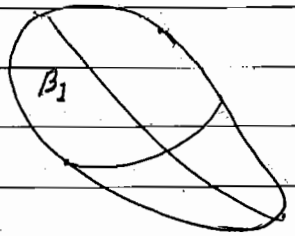
Que 1- Types, Distribution and functions of adrenergic reception ?

Types of adrenergic receptors.



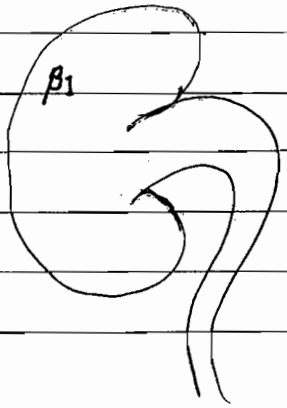
→ Distribution & Action :-

1.] Heart



- Force of contraction increase
- Heart Rate increase
- Conduction fast.
- Blood pressure increase.

2.] Kidney :-



- Increase release of renin
- Angeotensin increase
- Vaso constriction
- Blood pressure increase.

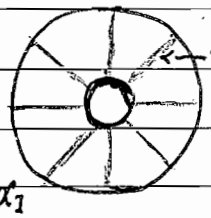
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3.] Vessels:-

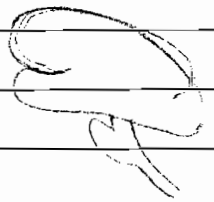
α_1, β_2

- α_1 Stimulated → Vasoconstriction → Blood Pressure increase
- β_2 Stimulated → Vasodilatation → Blood Pressure decrease
- Both stimulated :-
  - → Skin vessels → Vasoconstriction (more α_1) → Redistribution → blood to skeletal muscle
  - → Skeletal Muscle → Vasodilatation (more β_2)

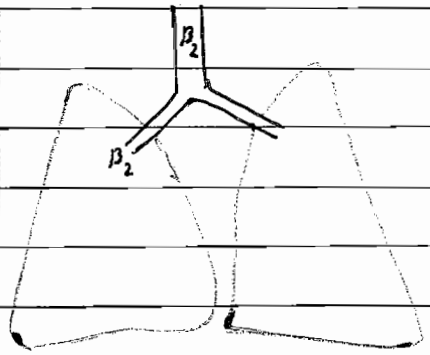
→ 4.] Iris:-



- Contraction of radial muscles.
- Mydriasis.

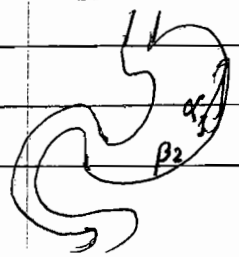


→ 5.] Respiratory:-



- Bronchodilatation.
- Secretions inhibite.
- Mast cell degradation stop

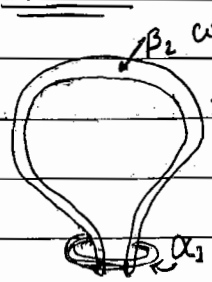
→ 6.] Stomach:-



- Decrease peristulsis movement
- GIT smooth muscle relaxation.

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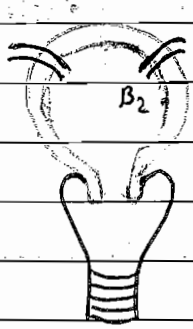
→ 7.] Bladder :-



- Bladder → Smooth muscle relaxation.
- Sphincter → Contraction
- Urinary Retention.

→ 8.] Pregnant Uterus :-

- Uterus Relaxation.



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→ 9.] CNS :-



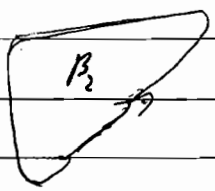
- α_2 presynaptic ⟶ Inhibitory action
- α_1 postsynaptic ⟶ Increase stimulatory action

→ 10.] Adipose Tissue :-



- Lipolysis.

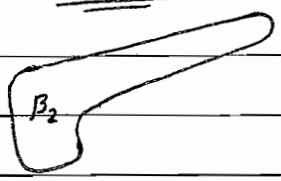
→ 11.] Liver :-



- Glycogenolysis
- Glucose synthesis.

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→ 12.] Pancreasel-



- Insuline Release Decrease
- Glucogon Release.

For Vivall-

→ IF → Ionotropic → Increase Force of contraction of Action cardiac muscle.

→ DC → Dronotropic → Increase Conduction velocity of heart Action

→ BF → Bethemotropic → Increase Exitability of Cardiac Action tissue.

→ CH → Chronotropic → Increase Heart Rate. Action

→ Adrenaline $\xrightarrow{\text{action}}$ $\alpha_1, \alpha_2, \beta_1, \beta_2, \beta_3$.

Noreadrenaline $\xrightarrow{\text{action}}$ $\alpha_1, \alpha_2, \beta_1$

Isoprenaline $\xrightarrow{\text{action}}$ $\beta_1, \beta_2, \beta_3$.

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☆ Classification of adrenergic Drugs?

ans i.] Drugs acting as CNS stimulant :-
- Amphethemine.
- Mephenthemine.

ans ii.] Drugs acting as Cardiac stimulant :-
- Adrenaline
- Dopamine
- Dobutamine.

ans iii.] Drugs used as pressure agent :-
- Noradrenaline
- Dopamine.

ans iv.] Drugs used as Bronchodilator :-
- Salbutamol
- Salmeterol
- Formeterol

ans v.] Drugs used as Uterine Relaxation :-
- Ritodrine
- Isoxsurpine
- Salbutamol

ans vi.] Drugs acting as Mydriatis :-
- Phynylephrine.

ans vii.] Drugs used as nasal decongestent :-
- Xylometazoline
- Phynylephrine.

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viii.] Drugs used as anorexient:-
- Sibutramine.

ix.] Local Vasoconstrictor:-
- Adrenaline.

✱ Uses of adrenergic Drugs? [A, B, C, D, E, G]

1.] Anaphylectic Shock:-
- It's type-1 hypersensitivity reaction.
- In which antibody are abnormally produced.
- They react with substance and mast cell degradation and vasodilatation occurs.
- Blood pressure falls down.
- Redness on skin.
- This condition is known as anaphylectic Shock.
- Adrenaline is used.
  - In IM dose (1:1000) 0.5 ml.
  - Adrenaline is physiological antagonist of Histamine.

2.] Bronchodilatation :- [Asthma]
- In asthma - Contriction of bronchus.
  - Increased fluid secretion.
  - Breathlessness occurs due to this.
- Adrenaline as subcutaneous or IV infution can be used.
- But its having more CVS stimulant action.

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- That's why
  Heart Rate increase
  Blood Pressure increase.

→ That's why more selective β_2 agonist are used.

- ex. → Salbutamol, Salmeterol, Formeterol.
- e.g. inhalation (use)
- So cardiac stimulation is prevented.

iii.] Cardiac Resuscitation :-

→ For cardiac arrest adrenalin is used.

→ Dose :-

1 : 1000 dilution → $\overrightarrow{IM}$ Use

1 : 10000 dilution → $\overrightarrow{IV}$ Use.

→ In cardiac arrest → Heart has stopped or decreased functioning.

→ Adrenaline stimulate β_1 receptor.

→ So, It increase force of contraction and increased conduction and increased heart rate.

iv.] Increasing the Duration of Local anesthetic.

→ Adrenaline is administered locally. it produce vasoconstriction.

→ So, Absorption of local anesthetic reduced.

→ and it's action is increased.

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v.] Epistaxis :-

→ Bleeding from the nose known as Epistaxis.

→ Adrenaline through α_1 receptor (nasal mucosa) produced vasoconstriction.

→ That will stop bleeding.

vi.] Glaucoma :-

→ Glaucoma is group of disorder characterised by increase in intraocular ~~pressure~~ tension and damage to the optic nerve & Ratina.

→ Adrenergic drug (adrenaline) acts on α_1 receptor and produce vaso constriction and decrease formation of aquous humour.

↓

decrease intraocular tention.

Que :- ★ Adverse Drug Reaction of Adrenaline ?

CVS :-
- Tachycardia.
- ~~Hypertension~~ - Hypertension.
- Palpitation.
- Cardiac Arrythmic.

CNS :-
- Insomia [Reduced Sleep]
- anorexia [decrease Food Intake]
- CNS Stimulant
- Tremor.

Smooth Muscle :-
- Urinary Retention
- Constipation.

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Que:- ☆ Uses of Adrenergic Drug? [5 marks]

☆ Dopamine?:-

→ It is directly acting catecolamine

→ It acts on α_1, α_2, β_1 and D_1 receptors. [Dopaminergic]

→ It is having dose dependant action.

• at Low Dose:-
- < 2 µg/kg/min.
• D_1 receptor stimulated which dilates coronary and renal arteries.
• So, Cardiac work is reduced peripheral resistance also reduced.

• at Moderate Dose:-
• 2 - 10 µg/kg/min.
• β_1 receptor is mainly stimulated which increase the force of contraction and increased heart rate.

• at Higher Dose:-
• > 10 Mg/kg/min.
• α_1 receptor is mainly stimulated,
• which produced vasoconstriction and increased peripheral resistance.
• Increase Noradrenaline releast → Blood Pressure increase.

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~> Use:-

-> That's why dopamine is used eg low or moderate dose.

-> [i.] In cardiac shock.

-> [ii.] Heart faillure with renal impairement.

-> [iii.] Dopamine cannot cross Blood Brain Barrier so, having no CNS effect.

[27-06-16]

Que:- Dobutamine ?

-> It is a catharcholamine having β_1, β_2 and α_1 action.

-> It is not acting on Dopamin receptor (D_1, D_2).

-> Most commonly Dobutamine act on β_1 receptor.

-> It increase the force of contraction of heart.

-> It increase heart rate.

-> So, it is used in CCF (Congestive Cardiac failure) and hypotension.

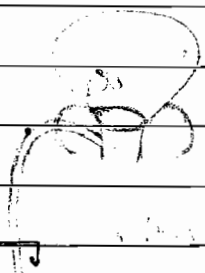
⇒ Adverse Drug Reaction:-

-> Tachycardia.

-> Hypertension.

Que:- α - Blockers ?

⇒ Classification



- Non Selective α Blockers
  - Reversible
    - Phentolamine
  - Irreversible
    - Phenoxybenzamine
- Selective α Blockers
  - α_1 Blockers
    - Prazosine
    - Tamsulosine
    - Terazosine
  - α_2 Blockers
    - Yohimbine

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⇒ Pharmacological Base:-

→ 1.] GIT:- - α - blockers
- Blocks the alpha receptors of GIT.
- and increase the peristalsis movement.
- Relaxes the sphincters.

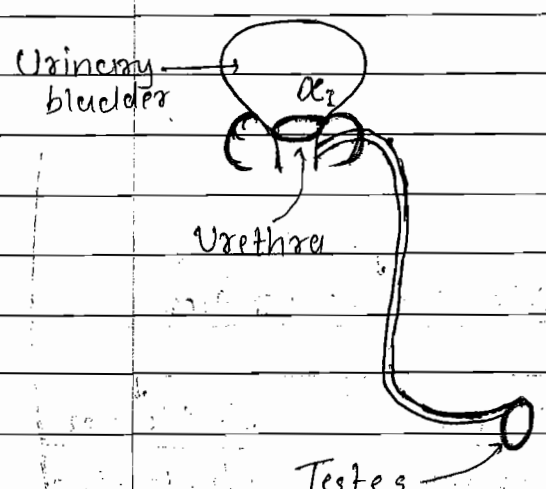
→ 2.] Blood Vessels:- - Blood vessels - dilated
- Peripheral Resistance decreass
- Blood pressure decrease.

→ 3.] Eye:- - Miosis
- Due to radial muscle - Relaxation.

→ 4.] Bladder:- - sphincter Relax
- Urine Pass.

→ 5.] CNS:- - Depression
- Reduced CNS activity.

→ 6.] Vasodiference:- → Retrograde ejeculation
- α_1 block
- Sphincter of bladder Relax
- Sperm is Retrogradly enters into bladder.
- Sexual disfunction occurs due to this.



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⇒ Uses:-

→ i.] Hypertension:-

→ alpha blockers used in hypertensive emergencies.

→ α_1 receptors produce vasoconstriction

→ α_2 Blockers, produce Vasodilatation and reduce peripheral resistance so reduce blood pressure

→ Hypertension emergency occurs in

[1.] Pheochromocytomas.

[2.] Clonidine withdrawal.

Note:-

→ Clonidine is α_2 agonist.

↓

acts on presynaptic neuron and reduce adrenergic action.

[Reduce Release of Noradrenaline].

So, used as antihypertensive.

→ ii.] Pheochromocytoma:-

→ It is the tumor of adrenal medula

→ So, increase production of adrenaline and Noradrenaline stimulation occurs.

→ For reducing the symptoms of ~~pheochromo~~ pheochromocytomas alpha blockers are used.

→ Definitive treatment is → Surgical removal of adrenal gland. (Tumor).

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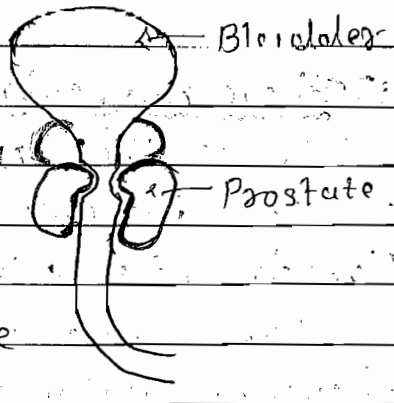
Benign Protate Hypoplasia (or)

⇒ [iii.] Benign Prostatic Hypertrophy (Hyperplasia) :- (BPH)

Hypertrophy:- Increase the size of cell.

Hyperplasia:- Increase the number of cell.

→ In this condition size of the prostate gland increase.
→ That compress urethra.
→ The flow of the urine is obstructed.
→ Frequency of urination increase.
→ Urgency will occur.
→ Incomplete passing of urine.
→ α - Blockers, Blocks the α receptor of sphincter and relax the sphincter.
→ So, Urine flow can be increased.
→ Selective α_1 Blockers are used.



- Along with α - Blocker, 5α reductase inhibitor is added
- To reduce the size of prostate.

⇒ [iv.] Peripheral Vasculer Diesease:-

→ In peripheral vasculer diesease there is vasoconstriction of peripheral vesiels.

→ α Blocker, block α_1 receptors of peripheral vessels
• and produce vasodilatation and reduce peripheral resistance.

⇒ [v.] Male Sexual Dysfunction :-

→ α Blockers used along with [papaverin] for erectile dysfunction.

⇝ Adverse Drug Reaction :-

→ Tachycardia
→ Postual hypotention
→ Nasal Stifness.
→ Importance due to retrograde ejeculation.

Que- ★ Beta Blockers ?

Non-Selective ($\beta_1 + \beta_2$)	Selective (Only β_1)
β - Blockers.	β - Blockers.
• Propranol	• Atenolol
• Timolol	• metoprolol
• Sotalol	• Bisoprolol
• Pindulol	• Esmolol

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Pharmacological action :-

→ Propranolol is studied as a prototype.

Mechanism of Action :-

→ It blocks β_1 and β_2 receptors competitive.

→ and reduce the sympathetic activity.

⇒ i.] CVS :-

- β blockers (propranolol) reduce the heart rate.
- Reduce the force of contraction of heart.
- Inhibits the activity of SA and AV node.
- Reduce the conduction of impulse.
- Reduce the Blood Pressure.

(Mainly due to decrease cardiac output).

⇒ ii.] Vascular System :-

- β-blockers produce vasoconstriction of peripheral vessels & Vessels supplying muscle.
- It may increase the blood pressure but long term therapy of β-blocker reduced the blood pressure mainly due to reduced the cardiac output.

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⇒ iii.) Kidney:-
- β_1 blocker reduce renin release from Ja apperatus.
- So, Angeotensin II Concentration is decreased Vasoconstriction is Reduced
- So, BP is reduced.

⇒ iv.) Respiratory System:-
- Non-Selective β-blocker.
- Block β_2 receptor of bronchial tree. (bronchu
- So, Bronchoconstriction occur.
- Asthma attack may occur.

⇒ v.) Skeletal Muscle:-
- It inhibits contraction of skeletal muscle.

⇒ vi.) CNS:-
- β-blocker supress the anxiety
- Produced the depression.

⇝ Metabolic Effects:-
→ β-blockers, reduce the warning symptoms of hypoglycaemia.

⇝ Uses:-
⇒ i.) Thyrotoxicosis:-
→ In thyrotoxicosis, thyroid hormone level is increase.
→ Hypertention, tremor, Anxiety is produced.
→ Non-selective β-blockers are used to reduce the symptoms of thyrotoxicosis.

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⇒ ii.) Esentical Tremor:-

→ Tremor produce due to excessive stimulation of β_2 receptor of muscle.

→ Non-Selective β-blocker is used to reduce these tremor.

⇒ iii.] Angina Pectoris:-

→ There is a chest pain due to blockage of coronary artery blood flow.

→ Selective β_1 blocker Reduce the cardiac output.

- Cardiac work is also reduced.
- So, demand of cardiac muscle is reduced.
- So, β blocker is cardiac protective.

⇒ iv.] Cardiac Arrythmia:-

→ Abnormal impulse generation in the conductive system of heart.

→ • Which produce abnormal production.

→ β-blocker reduce the conduction in heart.

→ So, arrythmia can be treated.

⇒ v.] CHF:- CCongestive Heart Faillure]:-

→ When the heart is not able to pump the blood according to the need of body is known as heart faillure.

→ So, the blood is accumulated in heart known as congetion.

→

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→ This is congestive cardiac feailure.
→ Selective β_2 blockers are used as chronic therapy (Meaintainance Therapy). in CHF beccause of it's cardioprotective action.

⇒ vi.] Hypertension :- [11-07-16]

Note:- Note!-

		Systolic	Diastolic
~~Note~~	Normal :-	120-139	80-89
Mild Hypertension	Grade-I :-	140-159	90-99
Moderate Hypertension	Grade-II :-	160-179	100-109
Severe Hypertension	Grade-III :-	≥180	≥110

→ Increased Blood Pressure.
→ Selective β_1-Blockers are used.
→ It reduce the force of contraction.
→ Reduce the conduction velocity.
→ Very effective in controlling Systolic blood pressur
→ Beccause of β_1-Blockers reduce renin secretion.
  • So, Reduce the effect of angiotensin-II
  • which helps in Reducing Blood Pressure.

⇒ vii.] Pheochromocytoma :-
→ It is the tumor of adrenal gland medulla.
→ Production of adrenaline and noradrenaline is increased.
→ So, Sympathetic stimulation occurs.
→ For Reducing the symptoms of α-Blocker + β-Blocker are used. (Non-Selectiv
→ Deffinative treatment is "Removal of Tumor".

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⇒ viii.] Glaucoma :-

→ Glaucoma is group of disorder characterised by damage to the Retina or optic nerve" but mostly it occurs due to increase intra occular pressure. (I.O.P.)

→ β- blockers reduce aqueous humor production

→ So, Intra Ocular Pressure is reduced

→ Timolol is used for this.

⇒ ix.] Myocardical Infuction :-

→ If there is reduced blood supply of myocardium, cells go into the stage of infuction (Cellular Death).

→ Selective β_1 - Blockers are cardio protective.

→ They are used as long term Regulation of Blood Pressure after myocardial Infuction.

• They also prevent future attack (Prevent Recurrance).

⇒ x.] Migrain :-

→ Headech which mostly occurs due to vasodilatation of the vessels of duramatter.

→ Non-Selective β- blocker, reduce, vasodilatation and prevent migrain.

⇒ xi.] Dissecting Aneurism :-

→ Aneurism means abnormal dilatation of artery.

→ When blood flows from the layers of artery known as Dissecting Aneurism.

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-> β- Blockers used for Reducing the chances of Rupture of artery by reducing B.P. and aortic pulsation.

Note:-

Cell Death

Necrosis	Apoptosis
• Localised cell death followed by inflammation.	• Programmed cell death.

[12-07-16]

Que:- Drugs for Glaucoma?

⇒ i.] β- Blockers :- Timolol ; Betaxolol.
⇒ ii.] Adrenergic Agonist :- Dipivetrin ; Brimonidine.
⇒ iii.] Prostaglandine analogue :- Latanoprost.
⇒ iv.] Carbonic anhydrase inhibitor :- Acetazolamide ; Dorzolamide.
⇒ v.] Cholinergic Drugs (Miotics) :- Pilocarpine.

-> First line Drug → Lumetoprost.
→ β- Blockers.

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☆ Autacoids ☆

Que :- Autacoids Instruction :-

→ "Self healing substances"

- Histamine,
- Serotonine,
- Prostaglandins,
- Leucotrians.

Que :- Histamine :-

→ Tissue amine.

→ Mainly ~~protect~~ produced by Mast cells.

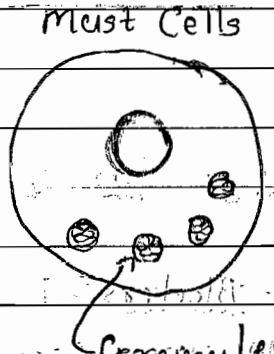
→ mast cells are granulocytes. (WBCs)

- Immature in circulation.
- They rest in tissue and become mature.
- They are seen in subcutaneous tissue, Lungs, GIT track.

Histidine (amino acid)

↓ decarboxylation.

Histamine



Que :- Classification of Antihistaminic Drug?

→ It is classified into two generations.

First Generation

- **Highly Sedative**
  - Hydroxyzine
  - ~~Hydroxyzine~~
  - Diphenhydramine
  - Promethazine
- **Moderately Sedative**
  - Phenaramine
  - Cinnarazine
- **Mid Sedative**
  - Chlorpheneramine

Second Generation

- Cetrizine
- Levocitrizine
- Azelastine

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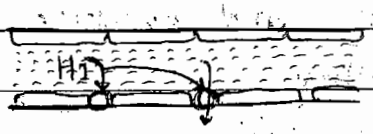
Que ✱ Pharmacological base of Antihistaminic Drug ?

→ There are 4 types of Histaminic Receptors.

H_1	H_2	H_3	H_4
• Subcutaneous tissue • Lungs	• GIT	• Presynaptic neuron	• Cells

⇒ H_1 Receptors :-

↝ i.] Skin :- Histamine → Increase vascular permiability and increas the fluid filtration.



→ Produce Oedema.
→ It increase inflammation.
→ Redness increase.
→ Stimulates nerve endings and pain and itching.

Antihistamine → Completly blocks histamine receptor.
→ and it reduce inflammation
. it reduce Oedema
. it reduce itching.
→ That's why antihistaminic is used in allergic reactions.

↝ ii.] Lungs :- Histamine → Increase the contraction of smooth muscle of trachea and bronchus.
→ It increase fluid accumulation.
→ So, it induce the attack of asthma.

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Antihistaminic → blocks the action of histamine and treats asthma.
(Receives the symptoms of asthma).
→ Most commonly Antihistaminic used in allergic asthma.
→ most cell stabiliser is better than antihistaminics.
→ Because they inhibite the release of histamine.

⇒ iii.] CNS:- Histamine → stimulate reticular formation and necessary for wakefullness, alertness.
Antihistaminic → Blocks this receptors and produce sedation and reduce irritability.
→ But this is minor action.
→ Because of extra anticholinergic action of antihismimic, they are good sedative.

→ First generation histaminic having anticholinergic action.
→ but second generation antihistaminic — having (No) anticholinergic action

Ques:- Difference between 1st generation and 2nd generation.

→ 1st generation	2nd Generation
- Having anticholinergic action	- No anticholinergic action.
- Produce sedation	- Can not produce sedation.
- Cross Blood Brain Barrier	- Can not cross BBB.
- Dryness of mouth, Constipation, urinary retention, Blurring of eye (Adverse drug reaction)	- Side effects are less.

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Que:- Uses of Antihistaminic Drugs?

⇒ i.] Allergy:-

→ Antihistaminic (H_1) competitively block h_1 receptor.
- So, it reduce permicability and fluid filtration into tissue.
- It reduce inflammation.
- Oedema is resolved.

→ So, antihistaminic used in the treatment of allergy.

→ e.g. Urticaria, Conjuctivitis, Allergic Asthma, Dermetiti Pruritus.

⇒ ii.] Vomiting:-

→ Histamin affects CTZ (Chemotriggyer Zone) and induce vomiting.

→ First generation antihistaminics can cross Blood Brain Barrier and block CNS, Histamine receptors and reduce vomiting.

⇒ Vertigo:-

→ Rotatory sense of oneself or surrounding is known as vertigo.

→ It occurs due to abnormal stimulation at vestibular apperatus.
- There are two types of receptors in vestibular apperatus (M, H).
- Antihistaminic blocks H_1 receptor and reduce the stimulation at vestibular apperatus and reduce vertigo.

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⇒ iv.] Common Cold :-

→ In common cold, there is increase of vascular permicability of nasal mucosa.

• So, increase fluid filtration.

• This occurs due to histamins.

→ Antihistaminic blocks, h_1 receptors and reduce Nasal dripping (Releives this symptoms).

⇒ V.] Hypnotics and Sedative :-

→ Antihistaminic (first generation) can cross Blood Brain Barrier and reduce irritability and conciousness.

→ So, they are used as sedative as well as hypnotic (for inducing sleep).

→ Histaminic do not produce dependance like benzodizepines.

⇒ vi.] Parkinson's Diesease :-

→ Parkinson's diesease is characterised by hypokinesia, rigidity and tremor.

→ Antihistaminics, along with centrally acting anticholinergic can reduce the symptoms.

→ It is used in Drug induce parkinsonism.

Psycosis	Parkinson
• Dopamin increase	• Dopamin decrease.

⇒ vii.] Preanesthetic Medication :-

→ Antihistaminic can be used before giving anesthetic drug.

→ Because it produce sedation and also reduce vomiting.

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Que:- Adverse Drug Reaction:-

⇒ First Generation:-
→ Dryness of mouth;
→ Urinary Retention;
→ Constipation;
→ Blurring of vision;
→ Sedation.

⇒ Second Generation:-
→ Contact Dermititis due to topical application.

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Que:- ★ Seratonin Receptors ?

Note

Production:-
→ Produced from tryptophan aminoacid.
→ It's hydroxylation and decarboxylation occurs.
→ also known as 5HT (Hydroxy tryptamine)

→ There are seven types of receptors.

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⇒ Receptor	Site	Action
→ $5HT_1$	• CNS	• Autoreceptors • Induce sleep • Works as neurotransmitter • Vasoconstriction of cerebral vessels.
→ $5HT_2$	• Platelates • Smooth muscles	• Platelate aggregation • Contraction of smooth muscle.
→ $5HT_3$	• CTZ	• Vomiting.
→ $5HT_4$	• GIT	• Peristalsis increase • Gastric emptying.
→ $5HT_{5-7}$	• CNS	

⇒ Vomiting treatment

Note

$5HT_3$ antagonist → Ondesetron

$5HT_4$ agonist → Metoclopramide.

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Ques Drug therapy for Migraine ?

=> Migraine is disorder characterised by throbbing headache, visual disturbance (with aura), photo phobia and vomitting.

=> Classification :-

Mild	Moderate	Severe
• One attack per month.	• 2 - 3 attack per month	• 2 - 3 attack per month
• Duration < 8 hrs. (It stays < 48 hrs)	• 6 - 24 hr (It stays 6-24 hrs)	• 6 - 48 hrs with visual disturbance (It stays 6 - 48 hrs)

-> Person is not able to perform routine work.

=> Drugs :-
-> NSAIDs
-> Antiemetic Drugs (Stop Vomitting)
-> Ergot Alkaloids.
-> Tryptans.

=> NSAIDs [Non Steroidal Antiinflammatory Drugs] :-
-> Paracetamol, Aspirin, Ibuprofen, Diclofenac, Naproxen.
-> They block the action of enzyme "Cyclooxygenase".
-> And reduce the production of inflammatory mediators, (PGs)
-> They reduce pain and inflammation.
-> These are used for mild to moderate migraine.
-> Long term therapy with these drugs are not advisable

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Antiemetic Drugs:-

→ Domperidone ; Ondensetron ; Metoclopramide.
→ They are used to revive the symptoms of vomiting.

Ergote Alkaloid :-

→ Ergotamine.
→ M.O.A.:-
- It is partial agonist at $5HT_1$ receptor.
- It induce vasoconstriction of cerebral vessels.
- It reduce the attack of migraine.
- It is used along with caffine.
- Caffine increase the action of ergotamine.

→ Dihydroergotamine :-
- It is prepared for parenteral administration.
(I.M., I.V., S.C. (Subcutaneous))

Tryptans :-

→ Sumatryptan
→ M.O.A. :-
- It is $5HT_1$ agonist.
- It produce vasoconstriction of cerebral vessels.
- and terminate the migraine attack.
- Mostly used in severe migraine.
→ Ergotalkloid and tryptan should not be combined.
→ It's action is fast and shorter acting.
→ ADR:-
→ Its cause hypertension so contraindicated in cardiovascular disorders.

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Que: Prophylaxis of Migraine?

→ When the person suffering from Moderate/Sever migraine, prophylaxis is used.

→ ~~Drugs used as prophylaxis of~~

⇒ Drugs used as prophylaxis of Migraine :- [A, B, C, D]

~ i.] Anticonvulsants:-
→ Sodium Valproate
• Can be used as prophylaxis of migraine.

Note:- When brain generates abnormal impulses, it produ convulsions.
⇒ That abnormal impulse generation is stopped by anticonvulsant drug.

~ ii.] Beta Blocker:-
• Propranolol can be used as prophylaxis of migraine.

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~ iii.] Calcium Channel Blocker:-
→ Verapamil
• Can also reduce the attack of migraine so used as prophylaxis.

~ iv.] Antidepressant:-
→ Tricyclic Antidepressant (TCA)
• Can be used as prophylaxis of migraine.

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Que:- Prostaglandins ? (Production and Action) ?

ans: Production:-

→ Membrane bound phospholipids.

↓

Arachidonic Acid

COX (Cyclooxygenase Enzyme) [COX_1, COX_2]

PgI_2	PgD_2	PgE_2	PgF_2	TXA_2	$PgF_{2\alpha}$
(Prostacycline)	Vasodilatation	Utrine Contraction	Utrine Contraction	[Thromboxine]	Decrease intra ocular pressure
Inhibite the platelate aggregation.		Gastric Protection		Aggregation of platelate	

ans Pharmacological Action:-

⇒ i.] GIT :-

→ Pg increase mucus protection.

→ It protect against acid.

→ Can be used in peptic ulcer.

⇒ ii.] Inflammation:-

→ Prostaglandins are responsible for inflammation

→ They stimulate nerve endings and produce pain.

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=> iii.] Uterus:-
-> Prostaglandins (PgE_2) increase the force of contraction of uterus.
-> It helps in a delivery process.
-> It prevents PPH (Post Partum Haemorrhage).
-> Used for abortion.

=> iv.] Eye:-
-> $PgF_{2\alpha}$ reduce the intraocular pressure.
-> So, used in the treatment of glaucoma.

=> v.] Blood:-
-> PgI_2 - Inhibite the platelete aggregation.
-> TxA_2 - Produce platelete aggregation.
-> Aspirine at low dose inhibite TXA_2 production.
So, used as antiplatelete drug.

=> vi.] CVS:-
-> PgD_2 - Vasodilatation and reduce peripheral resistance and reduce B.P.
-> PgE_1 - Helpful for opening of ductus arteriosus.

=> Uses of Pg:-
i.) Abortion
ii.) PPH (Post Partum Haemmorhage).
iii.) Induction of Labour.
iv.) Glaucoma
v.) Peptic Ulcer.

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Adverse Drug Reaction:-

→ Nausea.
→ Vomiting.
→ Hypertension
→ flushing.

Que:- Classification of NSAIDs:- [Nonsteroidal Antiinflamatory Drugs].

⇒ Nonselective NSAIDs:- (acting on COX_1 and COX_2)
- Aspirin
- Ibuprofen
- Meteneamic acid
- Indomethacin
- Piroxicam.

⇒ Preferential COX_2 inhibitor
- Diclofenac
- Aceclofenac

⇒ Highly Selective COX_2 inhibitor
- Etoricoxib
- Parecoxib

⇒ Drug with antipyrectic, analgestic and poor antiinflammatory action:-
- Paracetamol.

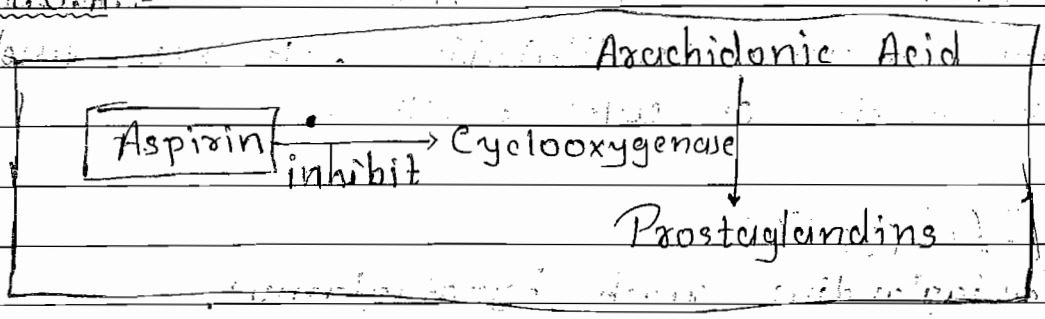
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Que:- Aspirin ?

→ It is non-selective COX inhibitor.

⇒ M.O.A.:-



→ COX (Cyclooxygenase Enzyme) present in Skin, GIT.
→ They are responsible for inflammation, pain and fever.
→ Aspirin inhibits COX enzyme and Reduce the production of prostaglandins.
→ and produce it's action.

⇒ Pharmacological Action:-

ans. 1.] Analgesic:-

→ Prostaglandins stimulate the nerve endings and produce pain.
→ Aspirin inhibit production of prostaglandins.
→ So, nerve stimulation reduced.
→ Pain is reduced [Analgestic].
→ So, it is used in headache, muscle pain, tooth pain, joint pain.
→ It's action is less than opoid.

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ans ii.] Antiinflammatory :-

→ Prostaglandins increase vasodilatation, Redness of site and chemotaxis of WBC

• So, inflammation increased.

→ Aspirin inhibit the production of prostaglandins and reduce vasodilatation, Redness and chemotaxis of WBC swelling.

ans iii.] Antipyretic :-

→ Prostaglandins reach hypothalamus.

→ and they effect thermostatic set point.

⇒ and increase the body temprature.

⇒ Fever Produced.

→ Aspirin inhibit prostaglandins and reset the body temprature.

• That's how it reduce fever.

ans iv.] Antiplatelate :-

→ Aspirin at low dose inhibite the production of TXA_2 [Thromboxine]

• So, platelate aggregation is reduced.

[50 mg - 325 mg].

ans v.] Kidney :-

→ Aspirin increase retention of sodium and water.

→ High dose ⟶ Increase erection of uric acid.

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vi.] GIT :-

→ Aspirin inhibitor COX enzyme of GIT.
• and reduce the production of mucus.
→ So, chances of peptic ulcer increased.

Note:-
- Aspirin is used along with proton pump. (omeprazole) inhibitor and the blocker (Ranitidine).
- So, chances of peptic ulcer decrease.

vii.] Uterus:-

→ Prostaglandins increase uterine contraction.
→ Aspirin inhibit production of Pg.
→ and delay labour.
→ Increase the chances of PPH [Post Partum Heremmorhaige]

viii.] Respiratory System:-

→ Aspirin – Increase chances of Respiratory alkalosis. [CO_2 washout]
- Followed by Metabolic acidosis

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⇒ Adverse Drug Reaction :-

→ Nausea, Vomitting.
→ Gastric, Peptic Ulcer.
→ Hypersensitivity reaction - rashes.
→ Respiratory alkalosis & metabolic acidosis.
→ Bleeding.
→ Rey's Syndrome :- In children suffering from hepatitis if aspirin used.
- The chances of liver failure increase [Encephalopathy].

⇒ Uses :-

→ Analgestic in headache, Toothpain, Joint pain, muscular pain.
→ Fever
→ Antiplatelate after myocardial infuction for reducing thrombosis.
→ Rheumatoid arthritis.
→ Gout.

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Que:- Salicylate Poisioning Or Overdose of Aspirin?

⇒ Clinicle Features:-
- → Nausica
- → Vomiting
- → Gastritis
- → Diarrhea
- → Fever
- → Acid-Base imbalance
  - hyperventilation (RR↑)
  - electrolyte disturbance (Oedema)
  - Concentration decrease
  - Convulsion
  - Coma → Death

Or,

⇒ Treatment:-
- → Hospitalisation
- → Gastric Lavage (with activated charcoal powder).
- → Injection of Sodium bicarbonate is given.
  - → M.O.A:-
    - Make urine alkali (Basic)
    - Aspirin is acidic, So, it is excreated more into basic alkali urine.
- → Fluid therapy
- → Electrolyte therapy
- → External Cooling
- → Hemodyalisis
- → Vitamin K.

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Ques: Uses of Other NSAIDs ?

⇒ i.) Ibuprofen :-
→ Muscular pain
→ In children espirin is contraindicated but ibuprofen can be used.

⇒ ii.) Diclofenac :-
→ It is concentrated in synovial fluid of joint.
• It is used in joint pain.

⇒ iii.) Indomethacin :-
→ It is used for closing the patent ductus arteriosus.
• also used as analgesic and antipyretic.

⇒ iv.) Mephenemic Acid :-
→ Dysmenorrhea (Painful Menstruation)
→ Analgesic and Antipyretic.

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Que: Difference between Selective COX_2 inhibitor and Non-selective COX inhibitor?

⇒ Selective COX2 inhibitor	Non-selective COX inhibitor
→ It has analgesic effect.	→ It has analgesic effect.
→ It has antipyretic effect	→ It has antipyretic effect.
→ It has antiinflammatory effect	→ It has antiinflammatory effect.
→ No antiplatelete effect	→ Having antiplatelete effect.
→ Not causing damage to gastric mucosa. • Not producing peptic ulcer	→ Causing damage to gastric mucosa, • and producing peptic ulc
→ These drugs increase the chances of thrombosis • So, increased chances of Cardiac damage.	→ Non-Selective drugs do not increase the chances of thrombosis or cardiac damag
→ e.g. Etoricoxip, Paracoxip	→ e.g. Aspirin, Ibuprophen, Mephanemic acid.

Que: ~~Aspirin~~ Paracetamol:-

⇒ M.O.A.:-

→ Inhibit cyclooxygenase enzyme and reduce the production of prostaglandins.
• It is good antipyretic and analgesic.
• But weak antiinflammatory.

⇒ Route:-

→ Oral or Parantral.

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⇒ Dose :-
→ 10 mg/kg [TDS]

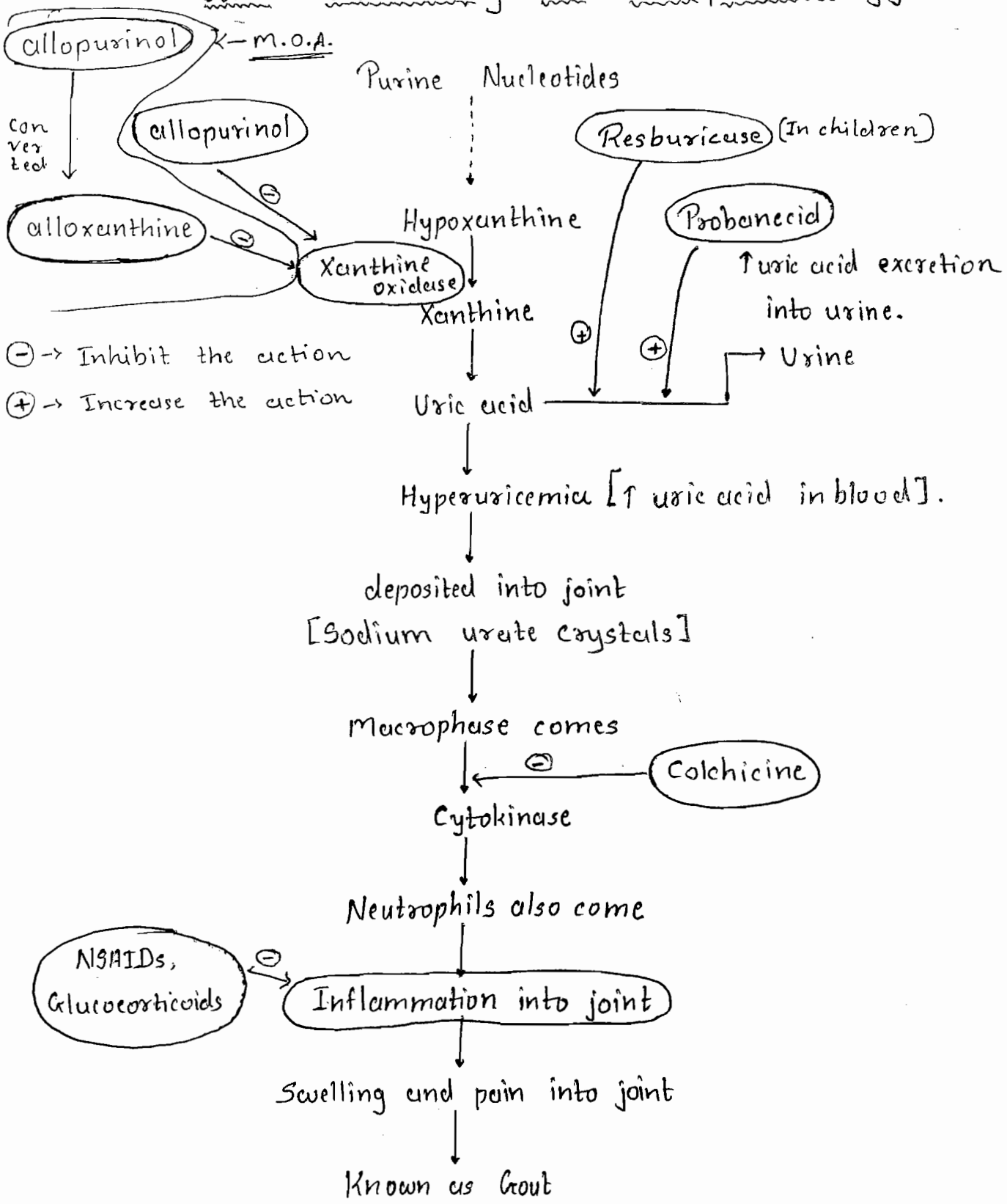
⇒ Use :-
→ i.) Analgesic :- headache, muscle pain, Toothache
→ ii.) Antipyretic :- for reducing fever
→ It can be used in child and pregnant women

⇒ ADR :-
→ Paracetamol is very safe drug.
→ Side effects rarely
• Due to chronic use :-
(i.) hepatic toxicity
(ii) Renal damage.

S.Q
Que: Paracetamol Poisioning :-

⇒ Antidote → N-acetyl cystine.

Gout biochemistry and Pathopharmacology.



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Classification of drugs used in Gout ?

There are two types of drugs used in Gout

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Que: Classification of drugs used in Gout?

⇒ There are two types of drugs used in Gout.

Acute Condition

i.] NSAIDs :-
- Diclofenac
- Indomethacine

ii.] Glucocorticoids :-
- Prednisolone
- Methylprednisolone
- Tricmeilone

iii.] Colchicine

Chronic Condition (for long term)

i.] Uricosuric Drugs.
- Probenecid

ii.] Inhibit uric acid production
- Allopurinol.

Que: Colchicine used in Gout?

→ It is an alkaloid.

⇒ M.O.A. :-

→ Colchicine inhibits the release of cytokinase (protein) from macrophase.

→ Inhibits chemotaxis of neutrophils.

→ and directly inhibits inflammation.

→ It is not acting directly for reducing pain.

⇒ Use :-

→ It is used in acute gout.

→ For long term use.

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⇒ ADR:-

-> Nausea, Vomitting, Abdominal pain.
-> Due to long term use - Bone marrow affected.
- Aplastic anemia.
- WBC reduced.

Que:- NSAIDs used in Gout?

-> NSAIDs inhibit cox enzyme.
·So, reduce prostaglandins production.
⇒ ·So, they reduce inflammation and swelling of joint.
-> Gives symptomatic Relief in the Gout.

Que:- Glucocorticoids used in Gout?

-> Glucocorticoids are strong antiinflammatory.
-> They inhibit inflammation and relieves the symptoms of gout.
-> Multiple joints affected → Give glucocorticoids via systemic route.
-> Single joint affected → Give glucocorticoid directly into joint (Intraarticular).

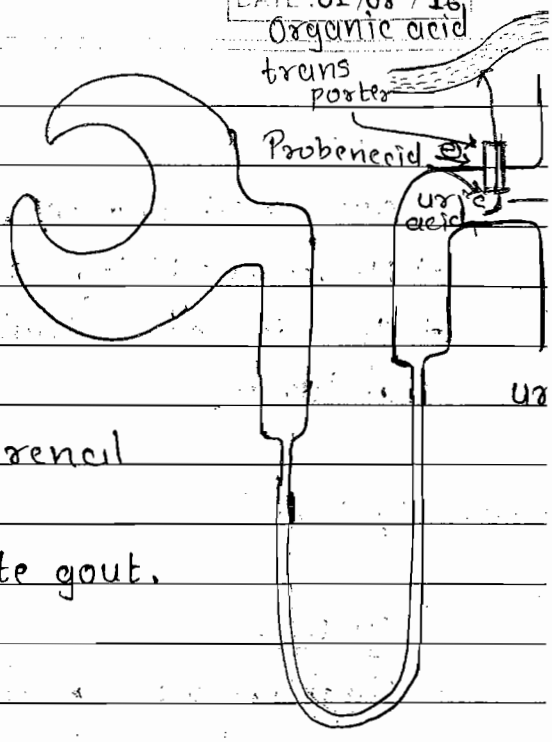
Que:- Probenecid used in Gout?

⇒ M.O.A.:-

-> Probenecid inhibit reabsorption of uric acid from renal tubules.
→ So, uric acid excreted more into the urine.

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Organic acid



=> Use:-
-> It is used in longterm treatment of gout.-

=> Care:-
-> It should not be used in renal failure.
-> It should be avoided in acute gout.

=> ADR:-
-> GIT disturbances
-> Abdominal pain
-> Rashes.

Que:- Allopurinol used in Gout?

=> M.O.A.:-
-> Allopurinol is converted into alloxanthine.
-> Alloxanthine inhibit enzyme [Xanthine Oxydase] (irriversibly) (Permanantly).
-> So, uric acid production is reduced
-> And it stops further progression of gout.

=> Use:-
-> It is used in early stage of gout.
-> It is used for long term therapy.
-> Used in secondary gout like gout due to renal failure or leucamia.

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=> ADR:-
-> GIT disturbance, Nausea, Vomitting
-> Hypersensitivity reaction -> rashes, itching
-> It is contraindicated in pregnancy, Lactation, children.

Que: Resburicase used in Children?

=> M.O.A.:-
-> Resburicase make uric acid soluble
. So, uric acid dilute in the urine and excreated via renal way or urination.

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★ Gastro Intestinal Track (GIT) ★

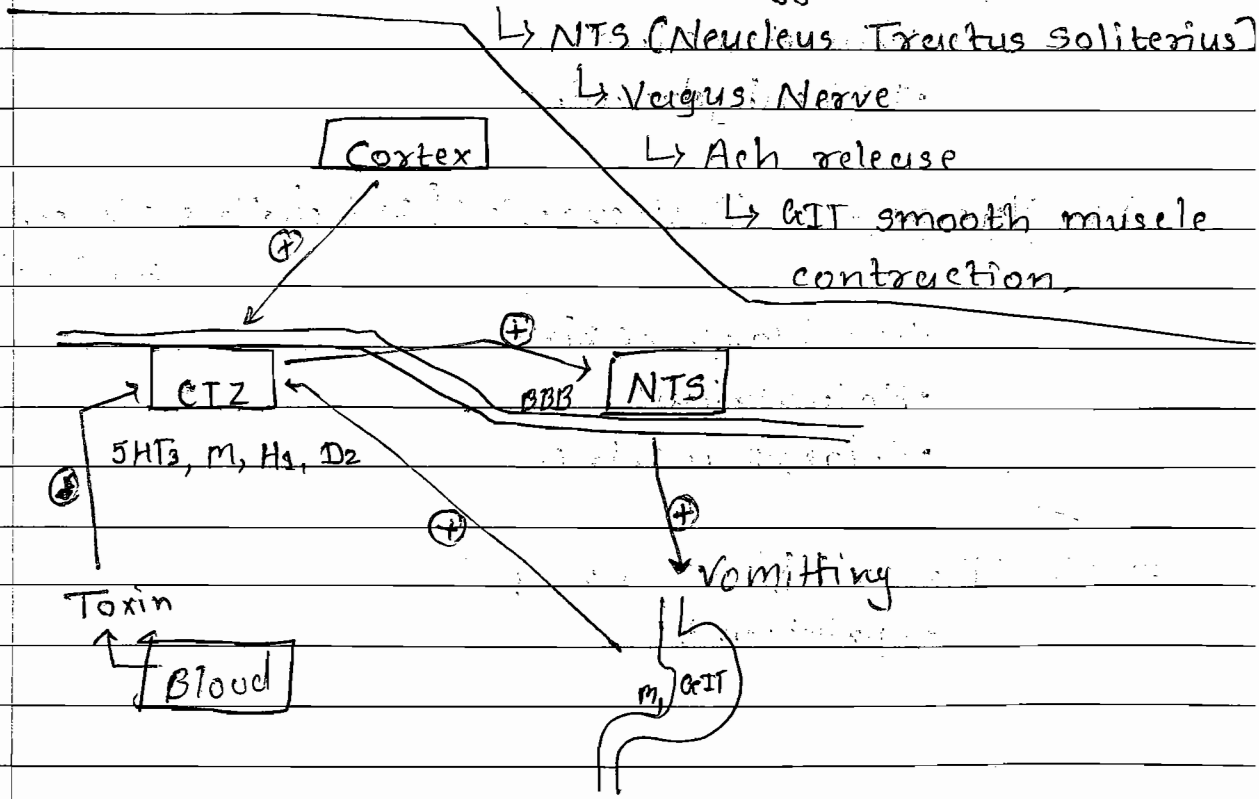
Que:- Mechanism of production of vomitting?

→ Vomitting is a protective reflex.
→ It is "reverse peristalsis" movement
→ It is necessary for ellimination of toxic substances present into GIT.

⇒ Reflex of vomitting :-

↝ Gensory - GIT mucosa
- Smell
- Vision
- Psycological factors.
- Taste.

↝ Nucleous :- Medulla oblongata
↳ Area postrema
↳ CTZ (Cemo trigger zone) (BBB absent)
↳ NTS (Neucleus Tractus Soliterius)
↳ Vagus Nerve
↳ Ach release
↳ GIT smooth musele contraction.



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Que:- Classification of drugs used to prevent vomitting ? (As antimetics)

=> i.] $5HT_3$ antagonist :-
- Ondensetrone
- Granisetrone
- Palonosetrone

=> ii.] Perokinetic Drugs :-
- Metochlopramide [$5HT_3$, D_2 Antagonist, $5HT_4$ agonist].
- Domperidone [D_2 antagonist].

=> iii.] Anticholinergic :-
- Hyoscine
- Dicyclomine

=> iv.] Antihistaminic :-
- Promethazine
- Diphenhydramine.

=> v.] Neuroleptic - (Antipsycotic) [Anticholinergic and Antihistaminic]
- Prochlorperazine
- Haloperidol
- Chlorpromazine.

=> vi.] NK_1 receptor antagonist :-
- Apripitant

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=> vii.] Dronabilol [Sedation]

=> viii.] Adjuvant Drugs :-

a.) Glucocorticoids :-

- dexamethazone
- Methylprednisolone

b.) Benzodizepenes :-

- Dizepam
- Lorazepam

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☆ 5HT3 antagonist ?

→ Ondesetron is prodrug.

⇒ M. O. A. :-

→ It blocks 5HT3 receptors.
→ It effects peripheral 5HT3 as well as central receptors.
→ by blocking 5HT3 peripheral receptors, it inhibits afferent impulse to CTZ.
→ by blocking central 5HT3 receptors, inhibit efferent impulse for vomitting

⇒ Pharmacokinetics:-

→ It is very well absorbed orally.
→ also effective via parentral route.
→ Transdermal patch is also available for long acting.

⇒ Uses:-

→ It is used for inhibiting the vomiting which is induced by chemotherapy or radiotherapy.
→ It is used for post operative vomiting.
→ It is used for pregnancy time vomiting

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=> ADR:-

->

-> This drug is very well tolerated and very less side effects.
-> Diarrhea.
-> Headache.

Que:- Metaclopramide:-

-> It is ~~~~ a prokinetic drug.

=> M.O.A.:-

-> It is $5HT_3$ antagonist.
-> It is $5HT_4$ agonist.
-> D_2 antagonist.

-> It is potent drug.
-> It can cross Blood Brain Barrier and also havin central effect.

=> Action is done by:-

-> increase lower esophageal sphincter tone.
-> relax pyloric sphincter.
-> increase gastric emptying.
-> increase peristalsis of small intestine.

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Prophylaxis :- Gluuia

⇒ Use :-

↝ i.] Postoperative Vomiting :-

→ Due to any surgical procedure vomiting is common.
→ So, as prophylaxis this drug can be used. (Treatment)

↝ ii.] Drug induced vomitting :-

→ Drug therapy most commonly associated with Nausea and vomiting.
→ This drug can be used as prophylaxis for preventing vomiting.
→ ex. antiparkinson's drugs "Levodopa" which increase chances of vomiting
→ Domperidon is prefered than metachlopramide.
• Metachlopramide can cross Blood Brain Barrier
• So, it inhibit central action of levodopa.
• Domperidon did not cross Blood Brain Barrier.
• So, it does not interfere with Levodopa action.

↝ iii.) GERD (Gastro Esophageal Reflex Disease) :-

→ In which content of stomach reflexly go back to esophagus and damage to esophagus.
→ Metachlopramide increase tone of lower esophageal sphincter and helps in releiving symptoms.

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-> PPI (Proton Pump Inhibitor) and H_2 blocker are prefered more than metaclopramide.

⇒ iv) Gastric Emptying :-

-> Metaclopramide relax pyloric sphincter.
-> and increase gastric emptying.
-> So, it is used before any operation for GIT emptying.

⇒ v.) Vomiting due to chemotherapy and Radiotherapy

-> Anticancer drugs induce vomiting.
-> So, metaclopramide is used for preventing chemotherapy induced vomiting.
-> Ondesetron is prepared over metoclopramide because of less side effects.

⇒ ADR :-

-> It can produce parkinsonism.
- Hypokinesia
- Muscle rigidity
- Tremor
- diarrhea
- Headache
- Galactorrhea
- Menstrual irregularities
- Gynecomastia

Note :-
- Dopamin inhibit prolactin Release
- Dopamin antagonist increase prolactin release.

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Que: Domperidon:-

→ It is a prokinetic drug.

⇒ M.O.A.:-

→ D_2 antagonist.
→ It is less potent than metachlopramide.
→ It can not cross Blood Brain Barrier.
→ So, central CNS side effects are not there.

⇒ Use:-

→ Drug induce the vomiting
ex used along with levodopa
→ Gastric emptying.

⇒ ADR:-

→ increase prolactin level
→ Galactorrhea
→ Menstrual irregularities
→ Gynaecomastia
→ Diarrhea

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Que: Anticholinergics are used as antiemetic?

→ Anticholinergic blocks muscaranic receptors of GIT track and inhibite abnormal contraction.
→ So, Hyocine and diclomine can be used as antiemetic.
→ It also blocks impulse from vestibular apperatus
→ So, used in motion induced vommiting.

Que: Antihistaminics used as antiemitics?

→ First generation H_1 blockers also having anticholinergic and sedative action.
→ So, it can stop vomiting.

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Que:- Mechanism of secretion of HCl ?

→ HCl is secreted by parital cells of gastric mucosa.
→ $H^+ - K^+$ ATPase pump is the final step for the acid secretion.
→ Muscarinic receptors in stomach stimulates acid production.
→ Histaminergic receptors (H_2) in stomach stimulates acid secretion.

⇒ **Prostaglandins :- (PgE_2) :-**

→ Increase the mucus production.
  • So, mucus protects stomach wall from HCl.
→ bacteria (*H. Pylori*) infects stomach and increase acid production.

Que:- Classify the drugs ~~and~~ used for peptic ulcer ?

⇒ i.] **Proton pump inhibitor :-**
[$H^+ - K^+$ ATPase antagonist]

- Omeprazole
- esmoprazole
- Pathoprazole
- Rabeprazole
- Lansoprazole

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=> ii.] H_2 blocker

- Renitidine
- Famotidine
- Cimetidine
- Roxatidine

=> iii.] Anticholinergic:-

- Pirenzepine

=> iv.) Prostaglandine analogue:-

- Misoprostole.

=> v.] Ulcer Protective:-

- Sucralfate
- CBS (Colloidal Bismuth Subcitrate)

=> vi.] Neutralise the acid secretion:-

(a) Locally acting:-

- Magnesium hydroxide.
- Alluminium hydroxide.
- Calcium Carborate.

(b) Systemically acting:-

- Sodium bicarbonate.
- Sodium citrate.

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=> vii.] Antibiotics (Antibacterial) drugs :-
- Amoxicilline
- metronidazole
- Tinidazole
- Clarithromycine

Que:- Proton Pump Inhibitor ? or Omeprazole Or PPI or $H^+ - K^+$ ATPase inhibitor ?

=> M.O.A.:-

-> It orally taken, reach to intestine than absorbed into the Blood, than reach to parital cell of stomach.
-> It converted into active metabolite (Sulfonamide)
-> That active metabolite binds irreversibly with $H^+ - K^+$ ATPase pump
-> final step of acid secretion is inhibited.
-> So, it reduce acid production and secretion.
-> It is also used intravenously.
-> It is taken befor meal (before 30 minutes)
-> Food interfears in its absorption.

=> Use:-

Peptic Ulcer:-

-> Demage to the GIT mucosa by increased HCl (acid) secretion produce ulcer, which is known as peptic ulcer.

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→ It is two type.
i.) Gastric Ulcer :- When occurs in stomach.
ii.) Duodenal Ulcer :- When occurs in duodenum.
→ Proton pump inhibitors reduced acid secretion and gives time for healing of ulcer.
→ Under due to H. pylori → PPI + Antibiotic used.
→ Stress ulcer → PPI are very effective.
→ H_2 blockers are also used.
→ Bleeding peptic ulcer → PPI are effective.

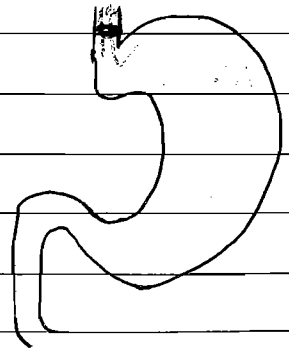
ii.) Zollinger Ellison Syndrome :-

→ It is due to "gastrin secreting tumour".
→ Gastrin production is high.
→ Gastrin stimulates acid production.
→ So, PPI are used to reduce acid secretion.
→ Permanent treatment is "Surgical removal of tumor".

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iii.) GERD :- (Gastro esophageal reflux disease) :-

→ Abnormal function of the lower oesophageal sphincter
→ Content of stomach reflex goes into the oesophagus.
→ Those content damage mucosa of oesophagus.
→ Pain is felt on epigestrium region.
→ PPIs are most commonly used drug in GERD.
→ because they are more potent than others.



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⇒ iv) Preoperatively :

-> PPIs reduce acid secretion.
-> So, they are used preoperatively used for reducing the chances of aspiration.

⇒ ADR:-
-> Diarrhea.
-> Abdominal discomfort.
-> Reduced absorption of vitamin B_{12}.
-> Reduced the absorption of minerals.

Note:-

Intrinsic Factor.

Produced from stomach — IF + Vit B_{12} — absorption

☆ H_2 blockers ?

⇒ Name:- Ranitidine
Famotidine
Cimetidine
Roxatidine

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=> M. O. A. :-

-> On parital cells of stomach, they block H2 receptors competitively.

-> Reduced acid secretion and production.

-> They reduced acid secretion during all the phases of gastric secretion. [Cephalic, Gastric, Intestinal].

-> They are less potent than PPIs.

-> They are most commonly used drugs.

-> Used orally as well as parentally

=> Uses:-

-> i.] Peptic ulcer:-

- Stress ulcer

↳ H2 blocker is most commonly used.

-> ii.] GERD

-> iii.] Zolinger Ellison's syndrome.

-> iv.] Preoperetivly.

=> ADR:-

-> Diarrhea

-> Abdominal discomfort.

-> Cimetidin can cross Blood Brain Barrier,

- So, it can cause confusion, Halucination, convulsion.

-> Cimetidine increase prolectin level so gynecomastia, menstrual irregulation.

-> So, cimetidine not used.

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Que ☆ Anticholinergic used in peptic ulcer?

→ Pirenzepine is an anticholinergic drug which block [muscarunic] [M] receptor and reduce acid production.

→ So, it can be used in peptic ulcer.

→ but it is not used routinely because it is less potent and having more side effects.

Que CBS used in peptic ulcer?
(Cadoinal Bismuth Subcitrate).

→ CBS (Bismuth)
- Pricipitant protein and helps into healing of ulcer.
- increase PG production, mucus production and bicarbonate production (Acid nutrition)
- Reduce the growth of H.pylori.
(Helicobacter pylori)

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Que: ☆ Anti H. pylori Drugs?

-> H. pylori is gram negetive bacteria, rod shaped
-> It resides into the mucosa of stomach.
-> It damage the mucosa and produce inflammation of mucosa.
-> So, it produce gastritis, peptic ulcer and gastric carcinoma.
-> Mechanism of production of damage →
  • Urease enzyme of bacteria produce ammonia and that ammonia may damage mucosa.
-> Combination of different drugs are used for the treatment of H. pylori infection.
-> Objective for combination of drugs:
  • For prevention of resistance.
  • For faster effect
  • For prevention of relapse.
  • For erradicate the H. pylori bacteria (Completely remove).

(or Virus)

-> Different combinations are discribed below:-

i.] Omeprazole / Lansoprazole
  Amoxicidine
  Clarithromycine.

ii.] Omeprazole
  Tetracycline
  metronedazole
  CBS (Bismuth)

-> Duration of treatment 2 weeks and PPI then continued for further 6 weeks.

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Quiz

Diarrhea :-

→ Loose stool passing 3 or more than 3 times in a 24 hrs

→ They are two types :-

Acute diarrhea	Chronic diarrhea.
- Mostly due to infection.	- When diarrhea occurs
- May be due to unknown etiology	for more than >2 wks. [AIDS]

⇒ Management of diarrhea :-

Or

S.Q] Anti-diarrheal drugs or Drugs used in diarrhea

i.] Non-specific therapy :-

a.] Rehydration therapy :-

- Oral rehydration therapy (ORS).
- Parantral therapy.

ii.] Antisecretory and antimotility drugs :-

a.] Opoids → Codeine, Loperamicle.

b.] α adrenergic agonist → Clonicline.

c.] Octreotide

d.] Racecadotril.

ii.] Specific treatment - Antimicrobials.

- Antibiotic - Ofloxasin, Ciprofloxacine.
- Antiparacytic - Albendazole.
- Antiamoebic - Metronidazole, Tinidazole, Omidazole.

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Que:- ORS (Oral Rehydration Solution) :-

→ It is used as Rehydration therapy for diarrhea.

⇒ Contents of ORS :-

→ Sodium Chloride	2.6 gm.
→ Pottasium Chloride	1.5 gm.
→ Sodium Citrate	2.9 gm.
→ Glucose	13.5 gm.

→ It is mixed with 1 litre of drinking water.

→ ORS is hypoosmolar 245 m.osm/L (milliosmolar)

	m.osm/L
⊕ Sodium	75
Glucose	75
⊖ Chloride	65
⊖ Citrate	10
Potassium	20
	245 m.osm/L.

⇒ M.O.A. :-

→ Solutes and water containing solution increase the absorption of contents from GIT in Blood.

→ It replenish the electrolytes into the body

→ and reduce the diarrhea by increasing absorption from GIT.

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-> It symptomatically treat the person.

=> Uses:-

-> Diarrhea
-> Dehydration [Mild to moderate]
-> Heat stroke
-> Burns.

-> Home base ORS prepration.
  - Boilled rice water.
  - Juice [Contining salt ang suger]
-> Recently WHO added Zinc [Zn] in to ORS.
  [10 - 14 days therapy].

Que:- ☆ Opoid anelouge for diarrhea?

=> 1.) Codeine:-

-> It is natural opoid anelougue.

-> M.O.A.:-
- Acts on μ receptor of GIT tract.
- and increases the tone of sphincter.
- so, sphincter are closed.
- decrease the GIT motility.
- So, diarrhea frequency is reduce.

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=> ii.] Loperamide :-

-> It is more potent opoid analougue for diarrhea.

-> M.O.A :-

-> Acts on μ receptor of GIT tract.
-> end: increases the tone of sphincter.
-> So, sphincters are closed.
-> Decrease the GIT motility
-> So, diarrhea frequency is reduced.

-> ADR :-
- Constipation
- Toxic megacolon
- Increased chances of perforation of GIT mucosa.
- Should not be used in children less than 4 yrs of age.

-> Uses :-
- In Traveller's Diarrhea.
- for decresing the frequency of diarrhea.

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Que. Racecadotrill :-

⇒ M.O.A. :-

→ It is elchephalinase inhibitor.
→ enchephalin elegradation is done by enchephalinase.
→ Racecadotrill inhibits enchaphalinase and increase concentration of enchaphaline.
→ It reduce secretion from intestinal mucosa.

⇒ Use :-

→ It secretory diarrhea (watery diarrhea).

Que. Octreotide :-

⇒ M.O.A. :-

→ It is somatostatin enalogue.
→ It inhibits the release of 5HT4 (Hydroxy treptamin)
→ So, secretion and movement of GIT tract reduced.

⇒ Use :-

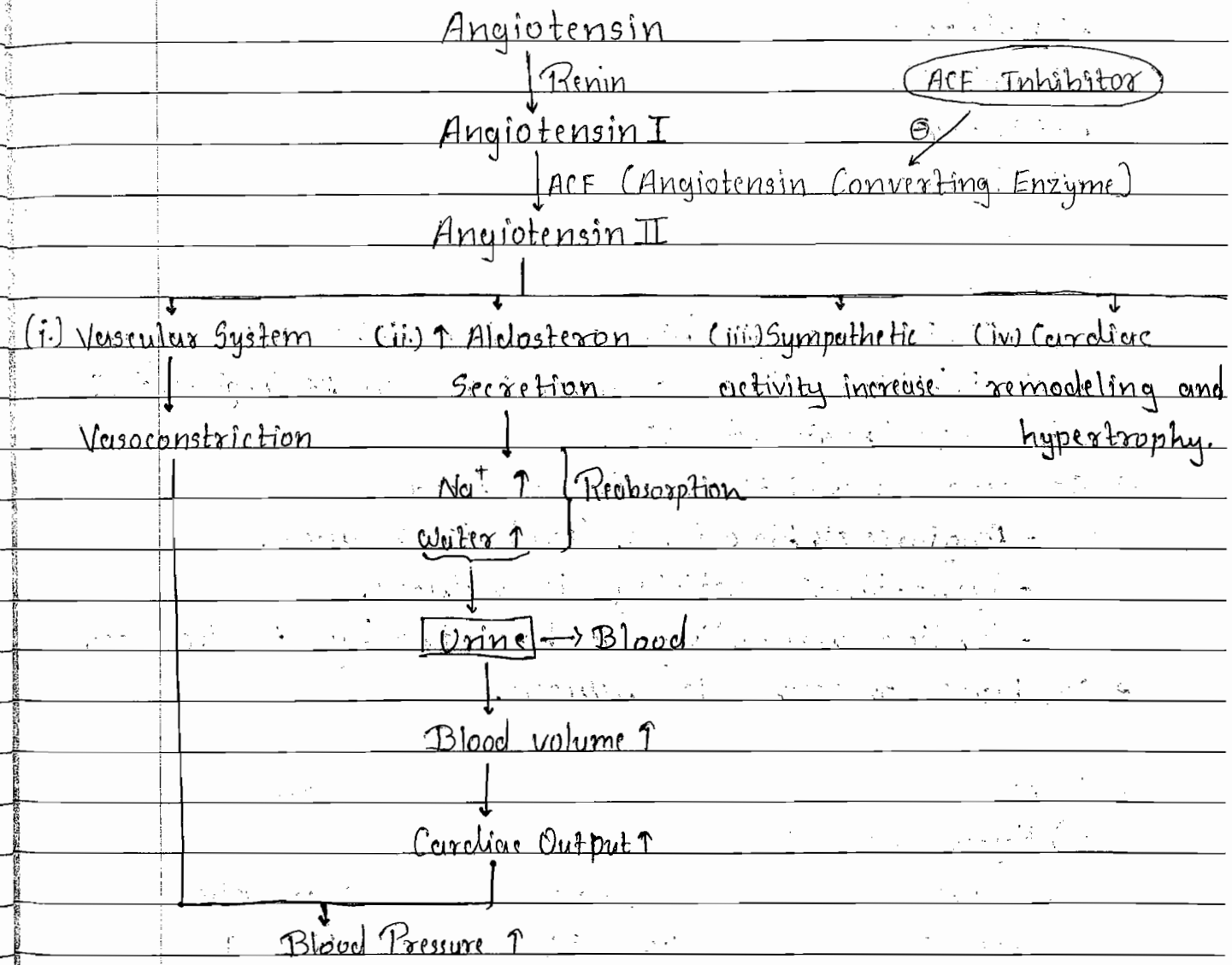
→ In the secretory diarrhea.

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☆ CVS (Cardio Vesicular System) :-

Que: Renin - Angiotensin Aldosteron System?

→ Renin is an enzyme secreted from JG apperatus.



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Que:- Write a short note on ACE inhibitor ?
[Name of ACE inhibitors, MOA, Uses, ADR]

→ Name of ACE inhibitor:-
- Captopril.
- Enalapril,
- Ramipril,
- Lisinopril,
- Perindopril.

=> m.o.a.:-
→ This drug binds with the ACE enzyme and inhibits the conversion of AT (Angiotensin I) into AII (Angiotensin II).
→ So, vasoconstriction is reduced
- ~~Angiotensin~~ Aldosteron level is reduced.
- Sympathetic activity is reduced.
- Cardiac remodling & hypertrophy is decreased.
→ So, blood pressure is reduced.

=> Use:-
→ i) Hypertension:-

Blood pressure	Systolic	Diastolic
- Stage I	140 - 159	90 - 99
- Stage II	160 - 179	100 - 109
- Stage III	≥ 180	≥ 110

→ ACE inhibitor reduce the level of angiotensin II into the blood.
→ So, vasoconstriction, sympathetic activity is reduced.

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→ So, BP is also reduced.
→ ACE inhibitors are used as oral or parentral medication.
→ They are used in all stages of hypertension.

→ ii.] Acute Myocardial Infarction:-
→ When blood supply to the cardiac wall is reduced, cells of myocardium go into the stage of Necrosis known as Myocardial infarction.
→ Cardiac remodeling occurs during Myocardial infarction.
→ Ace inhibitors inhibit cardiac remodeling and sympathetic stimulation.
→ So, ACE inhibitors are cardioprotective and used in myocardial infarction.

→ iii.] Congestive Cardiac Failure:-
→ When the heart is not able to pump enogh amount of blood known as cardiac failure.
→ And congestive changes occurs into the heart known as congestive cardiac failure.
→ ACE inhibitors prevent cardiac remodelling.
→ ↓ Blood volume → ↓ Cardiac Workload
→ So, ACE inhibitor used in CCF.

→ iv.) Diabetic Nephropathy:-
→ Due to diabetes chances of arthrosclerosis and Renal vascular and glomerular changes occurs.
→ ACE inhibitors produce dilatation of arteric and improve renal blood flow.

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=> ADR :- [C, APTOPRIL].

-> i.] Cough :- Bradykinin ↑
↓
Vasodilatation and affect Respiratory tract.

-> ii.] Angiodema :- Oedema at Lips, Larynx.

-> iii.] Proteinuriesis :- Occurs Rarely.

-> iv.] Teratogenicity :- Growth retardation of fetus, Renal damage, Renal death.

-> v.] Hypotension :- May occurs.

-> vi.] Neutropenia :- Rarely occurs.

-> viii.] Rashes

-> viii.] Itching

-> ix.] Loss of teste sensation.

-> x.] Hyperkalamia -> Increased K^+

=> It is contraindicated during Renal artery stenosis.

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Que:- Angeotensin Receptor Blockers (ARB) Name the drug, M.O.A., Uses and ADR ?

=> Name of ARBs:-
- Losartan
- Irbesartan
- Condesartan
- Telmisartan
- Valsartan
- Olmesartan.

=> M.O.A.:-

-> ARB competitivly blocks angiotensin receptors and antagonise the action of angiotensin-II.

Angiotensins are two types:
AT_1 ; AT_2.
-> ARB mainly binds with AT_1 receptors.
-> So, it inhibits the bineling of Angiotensin-II.
-> So, the action of angiotensin-II is reduced.
-> So, Hypertension is reduced.
Aldosteron secretion is reduced.
Sympathetic activity is reduced.

=> Uses:-
- i.] hypertension
ii.] Myocardial infarction
iii.] Chronic Cardiac Failure.
iv.] Diabetes Nephropathy.

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=> ADR:-
- Headach,
- Rashes, Itching,
- Hypotension,
- Teretogenicity,
- Hyperkalamia.

NOTE:-
- If the person suffers from cough due to ACE inhibitor than patient should be shifted on to the ARB.
- ARB do not affect ACE enzyme.
- So, metabolism of breadifinin is not effected.
- and cough is not produced.

Que:- Classification of antihypertensive Drugs?

=> i.] ACE inhibitor:-
- Ceptopril
- Enelapril
- Ramipril
- Lisinopril
- Perinopril.

=> ii.] ARB (Angiotensin Receptor Blocker):-
- Losartan
- Candesartan
- Telmisartan
- Velsartan
- Olmesartan.
- Irbesartan.

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=> iii.] Diuretics:-

1.] Thiazides:-
- Hydrochlorthiazide
- Chlorthalidone.
- Indepemide.

2.] Loop Diuretics:-
- Furosemide.

3.] K^+ Sparing Diuretics:-
- Spironolectone.

=> iv.] Calcium Channel Blockers:-

- Verapamil;
- Diltiazem;
- Nifedipine;
- Amlodipine.

=> v.) Sympatholytic Drugs:-

1.) β-blockers:-

a.) Selective β-blockers:-
- Atenolol
- Bisoprolol
- Mitoprolol
- Esmolol.

b.) Non-selective β-blockers:-
- Propranolol

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→ 2.) Centrally Acting :-
- Clonidine
- α - Methyl dopa.

→ 3.) β - blockers with additional α - blocking action :-
- Labetalol.
- Carvedilol.

Que:- Thiazides as Antihypertensive (Diuretics) (Name, Uses, ADR) :-

⇒ Name of Drugs :-

→ Thiazides → Hydrochlorthiazide
→ Chlorthalidone.

⇒ M.O.A. :-

→ Thiazides block $Na^+ - Cl^-$ Symport channel at the DCT of Nephron.
→ So, Na^+ is excreted in urine.
→ So, water is also excreted along with Na^+ ions.
→ Cardiac output is reduced.
→ So, B.P. is also reduced.
→ Due to Na^+ excretion peripheral resistance is also reduced.
→ B.P. is reduced.

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=> Advantages of Thiazides as antihypertension:-

-> Thiazides are well treated.
-> Effective in mild to moderate hypertension.
-> They are combined with other antihypertensive.
-> Cheap drugs.
-> Preffered in elder patients.
-> Longer duration of action.

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=> ADR:-
- Hypokalamia (↓ K^+ in blood);
- Hyperglycemia;
- Hypercalcemia;
- Impotence;
- Decreased Lipido (↓ Sexual Desire).

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Que:- CCB (Calcium Channel Blockers) as antihypertensive?

=> Names of CCB:-
- Verapamil
- Diltiazem
- Amlodipine
- Nifedipine

=> M.O.A.:-

-> Calcium Channel Blockers blocks calcium channel situated at smooth muscles of vessels and cardiac tissue.

=> Inhibit Calcium Current:-
-> Prevent the entry of calcium inside the cells.
-> So, smooth muscles contractibility is reduced.
-> Vasodilatation occurs.
-> Peripheral resistance reduce.
-> Blood Pressure is reduced.
-> Cardiac contractibility is also reduced.
-> So, Blood Pressure is reduced.
-> Verapamil and diltiazem reduce AV conduction.
-> So, heart rate is also reduced by them.

=> CCB is hypertensive:-

-> Very effective drug as monotherapy as well as combination with other.
-> Good for elder patiet.
-> It is prefered in patient suffering from asthma.

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Peripheral vascular diesease, Diabetes.

=> ADR :-

→ Nifedipin, Amlodipin } Reflex Tachycardia

→ Verapamil, Dilticazem } AV block.

→ Postrual hypertension, Headache, Oedema.

S.Q. Que: Amlodipine can be combined with β-blocker?

→ Amlodipine produce reflex tachycardia.
→ β-blockers can reduce tachycardia.
→ Both have different mechanism of action for reducing blood pressure.
→ So, they can be combined with antihypertensive effect.

S.Q. Que: Verapamil, dilticazem should not be combined with β-blocker?

→ Verapamil, dilticazem act on AV node and reduce conduction.
→ β-blocker also reduce conduction.
→ So, if they are combined person may suffer from heart attack.
→ So, they should not combined.

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Que:- β - blockers as Antihypertensive ?

=> Name of β - blockers :-

-> i.] Selective β_1 - blockers :-
- Atenolol
- Bisoprolol
- Mitoprolol
- Esmolol

-> ii.] Non - selective β blockers :-
- Propranolol.

=> M.O.A. :-

->

-> β - blockers blocks β_1 - receptor of heart
-> So, Cardiac Contractibility is reduced.
-> Cardiac output is also reduced.
-> So, blood pressure is reduced.
-> Non - selective β - blocker produce vasoconstriction So, selective β_1 blockers used more commonly.
-> β - blockers Reduce Renin Release.
-> So, conversion of angiotensinogen to angiotensin-I is reduced.
-> RAAS is suppresed.
-> So, Blood Pressure is controlled (Reduced).

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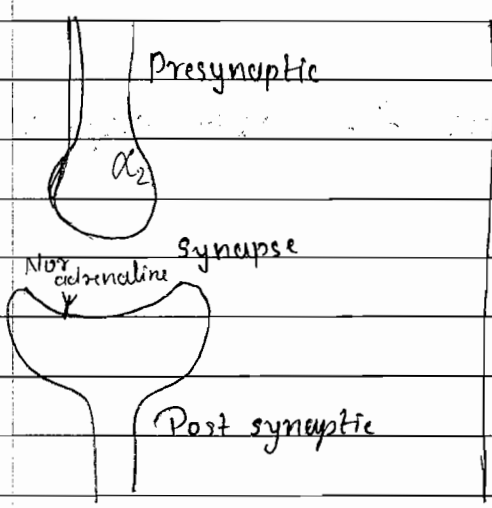
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⇒ β - blockers used as Antihypertensive :-

-> In young patient.
-> The patient having higher functioning of RAAS.
-> Patient having MI, Angina pectoris, other cardiac disorder.
-> Should not be used in Asthma patients.

Ques. Clonidine as Antihypertensive ?
S.Q.

-> Clonidine acts on α_2 receptors on presynaptic membrane of neuron.



-> Reduce the sympathetic outflow.

-> Reduce release of noradrenalin at synapse.

-> Blood Pressure is reduced.

-> It can cross Blood Brain Barrier.

⇒ Uses :-
-> Severe hypertension.

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Que:- Treatment of Hypertension?

→ Systolic B. P. ≥140 mm of Hg.
Diastolic B. P. ≥90 mm of Hg.
Considered as hypertension.

→ Stage of hypertension :-

B. P.	Systolic	Diastolic
Stage - I	140 - 159	90 - 99
Stage - II	160 - 179	100 - 109
Stage - III	≥180	≥110
Severe hypertension	>220	>120

⇒ Non-pharmacological treatment for early stage of hypertension:-
- Patient should be advised for
  - weight reduction.
  - Reduce NaCl diet.
  - Regular exercise
  - Less oil containing diet (Saturated food).
  - Stop smoking.

⇒ Pharmacological treatment for hypertension:-
- ACE inhibitor, ARB, β-blockers, calcium channel blockers, Diuretics are commonly used.
- They are used as monotherapy in early stage of hypertension.
- Combination therapy may be used for uncontrolled hypertension or severe hypertension.

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⇒ Drugs	Prefferred.
→ Hydrochlorthiazide (Diuretics)	- Mild Hypertension
→ Captopril, Enalapril, Ramipril or Losartan, Condesartan. (ACE, ARB)	- Mild to severe hypertensi - in young age. - [If person suffers from cough due to ACE inhibi then use ARB] - diabetic patient
→ Atenolol; Metoprolol (β-blocker)	- In young age. - if hypertension due to over sympathetic stimula - systolic hypertension.
→ Propranolol	- Contraindicated in asthma patient.
→ Amlodipine; Nifedipine (Calcium Channel Blockers)	• Old age • Hypertension due to PVD (Peripheral vascular diesease). • Mild to severe hypertensi

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=> Hypertension in Pregnancy:-

- α- methyldopa and niphedipine can be used in hypertension in pregnancy.

=> Therapy in severe hypertension:-

- Intravenous sodium nitroprusside can be used.
- Esmolol, Clonidine, Hydralazine, furosemide can be used.

→ 25 % of hypertension should be reduced first
→ B.P. → 160/100 mm of Hg should be maintained in 2-6 hrs.

Que:- Angina Pectoris:-
(Classification of drugs used in angina pectoris)?

→ Angina pectoris also known as chest pain.
→ There are three type of Angina Pectoris.

=> i.] Stable Angina:-

- In which angina pectoris occurs due to exersice of excessive work.

=> ii.] Unstable Angina:-

- In which angina occurs frequently and at resting stage also.

=> iii.] Varient Angina:-

- In which angina occurs due to severe vascular spasm.

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⇒ Pathology of Angina :-

↓ O_2 supply to the myocardium	↑ O_2 demand of body tissue
• Athrosclerosis	• increase heart rate
• Thrombosis	• Ventricular hyperthropy.

⇒ Treatment cupproch for cungical-

↑ O_2 supply to the myocardium	↓ O_2 demand
• PCI (Percuteaness coroneary intervension)	• Nitrates.
• CABG (Coroneary cirtery bypcass grafting)	• β-blockers.
• Nitrates - Antiplatelate drugs	• CCB.
- Thrombosis	

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⇒ Antianginal drugs :-

→ 1.] Nitrates :-

a.) Short acting :-
- GTN (Glyceryl trinitrate).

b.) Long acting :-
- Isosorbide dinitrate.
- Isosorbide mononitrate.

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→ 2.] β - blockers:-
- Atenolol ; Metoprolol ; Esmolol ; Bisoprolol.

→ 3.] Calcium Channel Blockers:-
- Verapamil
- Diltiazem
- Amlodipin
- Nifedipine

→ 4.] K^+ Channel Blocker:-
- Nicorandil

→ 5.] Other Drugs:-
↳ Antiplatelate Drugs:-
- aspirin ; clopidogrel.
↳ Statins
- Atrovastatin.

Que:- M.O.A. ; Pharmacological base ; Use ; A.D.R. of Nitrates?

⇒ Nitrates:-
i.) Short Acting:-
- GTN
ii.) Long Acting:-
- Isosorbide dinitrate.
- Isosorbide mononitrate.

⇒ M.O.A. :-

- Nitrates produce Nitrogen Oxide.
- It increase the activity Guanyl Cyclase.
- So, cAMP concentration increased.
- Camp (Cyclic cAMP) decrease Ca^{2+} Concentration in the cell.
- So, it inhibits muscular contraction. (Smooth).
- Vasodilatation occurs.

⇒ Pharmacological base of Action:-

≈ 1.] Vein :-

- Nitrates produce NO (nitrogen oxide).
- So, it relax smooth muscles of vein.
- Venodilatation occurs.
- Venous return to the heart is reduced.
- So, cardiac output and workload is reduced.
- So, Anginal pain is reduced.

≈ 2.] Artery :-

- Nitrates produce arteriodilatation.
- so, peripheral resistance is reduced.
- So, cardiac workload is reduced.
- Angina pain can be reduced.

≈ 3.) Heart (Coronary Vessels) :-

→ Nitrates increase the blood flow towards ischemic area of heart.

→ It known as the redistribution of blood.

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=> Uses:-

1.] Angina Pectoris:-

- Due to decreassed blood supply to the myocardium angina pectoris occurs.
- GTN is used in acute attack of angina
- It is given ~~in~~ sublingually because it is faster absorbed and bypass the first pass metabolism.
- If pain is releived and person starts developing headache than person should ~~split~~ spit out tablet.
- Maximum 3 tablets can be used at a time.
- Long acting nitrates are used for prophyluxis of angina.
- Prophyluxis should be used in unstable angina.
- Due to long term treatment with nitrates, Tolerance developes against it's action.

2.] Myocardial Infaction:-

- Infaction of myocardium due to reduced blood supply to myocardium.
- Nitrates can be used in acute attack of myocardial infaction.
- but it alone can not reduce the pain of myocardial infaction.
- Blood pressure should be monitered during nitrate theruphy.
- should be stopped if hypotension occurs.

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3.) CCF :- (CHF):

- If heart is not able to pump able to pump according to demand of body tissue known as heart failure.
- Blood is accumulated in heart chamber known as CCF.
- Nitrates reduced pre-load and afterload so, cardiac workload is reduced.
- B.P. monitering should be done.

4.) Hypertensive Emergencies :-

- Intravenous nitrates can be used in the treatment of hypertension.

5.) Billiary Colic :-

- Due to obstruction of bile duct ; pain occur known as billiary colic.
- Nitrates relax bile ducts and releives pain.

6.) Cynide Poisioning :-

- Cynide inhibits cytochrome enzyme and inhibits cellular metabolism.
- MOA of nitrates in cynide poisioning.

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→ Hemoglobin —(Nitrites, +)→ Methhemoglobin

+

Cynide

↓

I.V. { Cynomethhemoglobin + Sodium thiosulphate

↓

Sodium thiocynate

↓

Excreted in Urine.

⇒ A.D.R. :-

- Headach
- Hypotension
- Tachycardia
- Palpitation
- Methhemoglobinemia.

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☆Respiratory System☆

Ques:- What is cough ? Types of cough.

→ Cough is a protective reflex for removal of substances from the Respiratory tract.

⇒ Types:-

→ i.) Productive Cough:-
- Cough with substances coming out known as productive cough.

→ ii.) Non-productive Cough:-
- Cough without sputum (substance) known as non-productive cough.

⇒ Mechanism of production of cough:-

→ i.) Local irritation:-
- Irritation to the respiratory tract mucosa produce the cough.

→ ii.) Central Regulation:-
- Cough centre is located into the medull oblongata.
- When the centralis stimulated it can produce cough.

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Que:- Classification of drug used in treatment of cough?

⇒ 1.] Pharyngeal demulsants:-
- Lozenges.
- Linctuse.

⇒ 2.] Antitussive:-

↝ i.) Opioid Antitussive:-
- Codeine.
- Pholcodeine.

↝ ii.] Non-Opioid Antitussive:-
- Noscapine.
- Dextromethorphane.

↝ iii.] Antihistaminic:-
- Chlorpromazine.
- Promethazine.

⇒ 3.] Mucolytics:-
- Bromhexine.
- Ambroxole.

⇒ 4.] Expectorants:- [Secretion Enhancer]:-
- Sodium citrate.
- Potessium citrate.
- Guaphenesine.
- Ammonium chloride.
- Balsum of tolu.

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Que- Describe M.O.A., Pharmacological base, ADR of antitussive?

⇒ Name of antitussive drugs :-

↝ Opioid :- Codeine;
Pholcodeine

↝ Nonopioid :- Noscapine;
Dextromethorphane

↝ Antihistaminies :- Chlorpromazine;
Promethazine.

⇒ M.O.A. :-

→ These drugs can act on central cough centre.
→ They supress the cough centre.
→ So, cough can be stopped.

⇒ Pharmacological base with ADR :-

↝ Opioid :-

- These drugs are very potent in suprising cough.
- ~~Should~~ should be used in non productive cough.

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-> ADR :-
- It inhibits intestinal movement and produce constipation.
- It produce dependence.
- It produce drowsiness.
- Bronchospasm.

=> Non Opioids :-
- This drugs also acts similar to the opioids.
- But side effects like constipation and dependence is less.

-> ADR :-
- Headache
- Bronchospasm.

=> Antihistaminics :-
- 1st generation antihistaminics which are also having anticholinergic action.
- They reduce the cough and reduce secretion.
- They are good for cinergic type of cough.

-> ADR :-
- Dryness of mouth.

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Que:- Classification of drugs used in asthma ?
or
Antiasthmatic drugs ?

=> Defination:-
Asthma:-
- Hyperresponssiveness of trachio-bronchial tree, which results in broncho-constriction and difficulty in breathing known as asthma.

=> Types of asthma:-
i.) Acute asthma.
ii.) Chronic asthma
iii.) Status asthmeticus.
-(Continuous attack of asthma).

=> Classification of drugs used in asthma:-

~> 1.] Bronchodilators:-
i.] β_2 - agonist selective:-
- Salbutamol
- Salmeterol
- Formeterol
- Terbutaline

Non-Selective agonist (Sympathomimetics)
- Adrenaline.

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ii.] Methylzenthine:-
- Thiophyline
- Aminophyline

iii.] Anticholinergic:-
- Ipratropium bromide
- Tiotropium bromide.

=> 2.] Mast Cell Stabilliser:-
- Sodium chromoglycate.
- Ketotifen.

=> 3.) (Leukotriens) LT antagonist:-
- Monteleukast.
- Zafirleukast.

=> 4.) Glycocorticoids:-
-> i.) Inhaled Steroids:-
- Beclomethasone
- Budesonide
- Fluticasone
- Cyclesonide

-> ii.) Systemic Steroids:-
- Hydrocortisone
- Prednisolone
- Methylprednisolone.

=> 5.) Anti Ag E antibody:-
- Omelizumeb.

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Ques β - agonist in asthma?

=> Name of β - agonist used in asthma.

- Salbutamol
- Salmeterol
- Formeterol
- Terbutaline.

=> Non - selective

- Adrenaline

=> M.O.A. :-

-> It acts on the smooth muscle of trachio-bronchial tree.
-> On the β_2 receptors.
-> Increase cAMP concentration inside cell.
-> Smooth muscle of trachio-bronchial tree relax.
-> Bronchodilatation occurs.

minor action
-> also acts on mast cells.
-> Reduce the action of inflammatory mediator.

=> Description of β_2 agonist:-

ans i.] Salbutamol :-

-> Onset of action -> fast.
-> Duration of action -> short (upto 6 hrs).
-> So, used in the acute therapy for asthma.
-> Due to short duration of action, it is not used for chronic asthma.

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ans ii.) Salmeterol:-

- Onset of action → slow
- Duration of action → Long (Upto 12 hr)
- So, it is used in chronic therapy of asthma.
- Not used in acute attack of asthma.

ans iii.) Formeterol:-

- Onset of action → fast
- Duration of action → long
- So, it can be used in acute asthma and as chronic therapy of asthma.

⇒ ADR:-

- Tremor.
- Tachycardia.
- Arrythmia.
- Palpitation.

Que: Methylxanthines (Phosphodiesterase inhibitor):-

⇒ Name of methylxanthine:-

- i.) Theophylline
- ii.) Aminophylline
- iii.) Etophylline
- iv.) Doxophylline

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=> M.O.A.:-

- Methylxanthines are inhibitors of phosphodiesterase enzyme.
- Function of phosphodiesterase enzyme (PDE) is to metabolise cAMP.
- Methylxanthines inhibits PDE and increase cAMP.
- So, smooth muscle relaxation occurs.
- Bronchodilatation occurs.

=> Use:-

-> Thiophyline → Used orally
→ as poor water soluble, IV not used.

-> Aminophyline → Orally used.
→ as irritant, it should by used slowly for IV therapy.

-> Methylxanthines are used in asthma and COPD patients.

-> Due to more side effects, their use is limited.

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⇒ ADR:-

→ On CNS:-

- Restlessness, insomia, convulsion.

→ On CVS:-

- Tachycardia; arrythmia; Hypotension.

→ On GIT:-

- Nausea; vomiting; Gastric & peptic ulcer.

→ On Kidney:-

- Diuresis.

→ ADR according to dose of each drug.

Que: Anticholinergic in Asthma?

⇒ Name of drugs:-

- Ipratropium bromide
- Tiotropium bromide.

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⇒ M.O.A.:-

- These drugs blocks (antagonise) the muscranic receptors of tracheobronchial tree.
- So, produce bronchodilatation.

⇒ Use:-

- It is less potent than β - agonist.
- It is used in acute attack of asthma along with β - agonist.
- Combination therapy of anticholinergic with β - agonist is more effective.
- Used via inhalation route.

⇒ ADR:-

- Dryness of mouth.
- Tachycardia.

Que:- Mast Cell Stabilizer?

⇒ Name of Drugs:-

- Sodium Chromoglycate.
- Ketotifen.

⇒ M.O.A.:-

- This drug stabilizes the mast cell membrane and prevents the degradation of mast cell.
- So, release of histamine, PG (Prostaglandins) leucotriens are reduced.
- So, asthma attack is prevented.

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=> Use:-

i.] Allergic Asthma:-

- In allergic reaction mast cells play key role for production of asthma.
- So, mast cell stabiliser can be used.
- It is not effected orally.

ii.] Allergic conjuctivitis
iii.] Allergic rhinitis
iv.] Allergic dermitis

=> ADR:-

- Local irritation.
- Cough
- Headache.

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Que:- Leukotriens Antagonist ?

=> Name of Drugs:-
- Monteleukast
- Zafirlukast

=> M.O.A.:-
- These drugs blocks LT1 receptors of smooth muscles of trechiobronchical tree.
- So, Leucotriens which are inflammatory mediater not able to bind with LT1 receptor.
- So, bronchodilator is produced, inflammation is reduced and asthma is treated.

=> Use:-
- Used in moderate to severe asthma.
- Orally well absorbed and plasma protein binding is more.
- Preffered in children.
- They are well tolerated.

=> ADR:-
- Headach
- Rashes
- ~~Eosin~~ Eosinophillia.

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Que:- Steroids in asthma?

→ Name of the glucocorticoids used in asthma.

~» Inhaled Steroids:-
- Beclomethasone.
- Budesonide.
- Ciclesonide.
- Fluticasone.

~» Systemic Steroids:-
- Hydrocortisone.
- Prednisolone.
- Methyl Prednisolone.

⇒ M.O.A.:-
- Steroids inhibit phospholipase enzyme and reduce production of inflammation mediator.
  - So, steroids [Reduce inflammation.]

- Steroids prevents antigen - antibody production.
  - So, it works as [anti allergic.]

- Reduce antibody production.
  - So, it works as [Immunosupresent.]

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⇒ Uses:-

i.] For moderate asthma:-

- When asthma attack is > 1 per day known as moderate dose.
- Inhaled steroid with long acting β_2 agonist, is used.
- Steroids prevents the tolerance of β_2 agonist.

ii.] For Severe Asthma:-

- Contineous attack of asthma for long duration is known as status asthmeticus.
- Intravenous systemic steroids are used for the treatment.

iii.] For Status Asthmeticus:-

- Contineous attack of asthma for long duration is known as status asthmeticus.
- Intravenous systemic steroids are used for the treatment.

⇒ ADR:-

- Dryness of mouth
- Fungal infection of oral cavity.
- Supression of hypophyseal - pituitary axis.
- Osteoporosis.
- Hypertension.
- Muscle weakness.

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⇝ For prevention of ADR of steroids.
- Rinsing (Wash) the mouth after each inhaled dose.
- Use the spacer.

Ques- Status Asthmatics :-

→ Contineous attack of asthma for long time is known as status asthmitis.

→ It is associated with upper respiratory track infection.

⇒ Therapy :-
- Start humidified O_2.
- Give intravenous hydrocortisone 100 mg stat followed by 100 mg 4-8 hrly.
- Give β_2 agonist (Salbutamol) (faster acting) by Nebuliser.
- Start $\vec{IV}$ fluid therapy.
- Start appropriate antibiotic therapy.
- If patient do not responde to the Nebuliser therapy.
  - than intubation is done and bronchodilators are given.
- Tracheostomy is done if intubation is difficult.
  - Mechanicle ventilation is done.
- If person suffers from hypokalemia potassium therapy is given.
- If acidosis occurs → $\vec{IV}$ Sodium bicarbonate.

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★ BLOOD ★

Que:- Coagulation:-

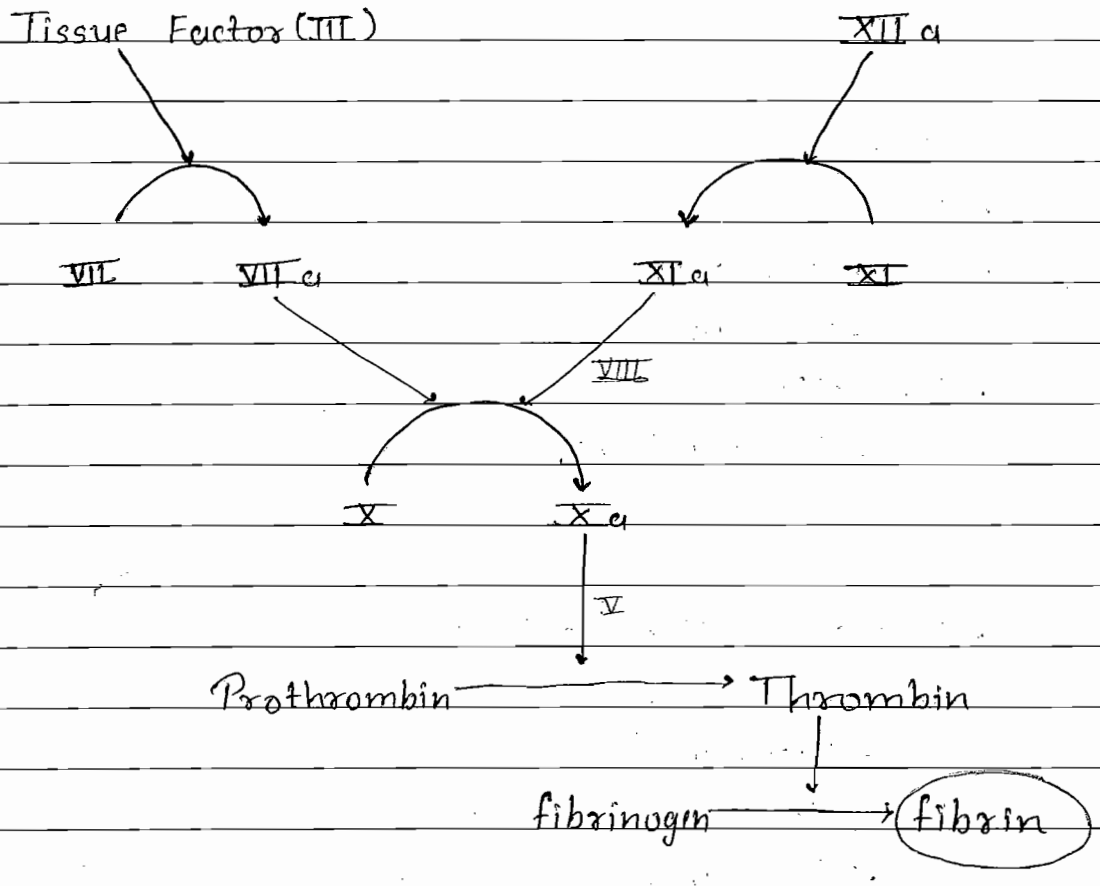
→ It is the process of conversion of fibrinog into fibrin and formation of clot.

→ Pathway of coagulation:-

- There are two pathways of coagulation:

Extrinsic Pathway

Intrinsic Pathway



Insoluble clot formation.

Vitro:- Outside of body Vivo:- Inside of body

Que:- Classification of anticoagulant drug?

Ans 1.] Anticoagulant used In vitro:-

- Heparin
- Sodium Citrate
- Sodium EDTA [Ethyline Diamide Tetra Acitate]
- Sodium Oxalate

Ans 2.] Anticoagulant used In Vivo:-

⇒ i.] Parentral:-

- Heparin (UFH → Unfractioned heparin).
- LMWH (Low Molecular Weight Heparin).
  - enoxaparin.
  - ravipurin.
- Synthetic
  - fondaparinux.

⇒ ii.] Oral:-

- Coumarin DeriNative:-
  - Warfarin.
  - Dicumarol.
- Inendione Derivative:-
  - Phenindione.
- Factor Xa Inhibitor:-
  - Rivaroxeiban.

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Quer- M.O.A. ; Uses and ADR of heparin ?

→ Heparin is mucopolysacharide.
→ It is negetivly charged organic acid.
→ It is plasma protein bound.
→ It is normally present into human body.
→ It is very large molecule.

⇒ M.O.A. :-

→ Heparin indirectly inhibit coagulation.
→ Heparin activates antithrombin
- and that ~~antithrombin~~ anti-thrombin inactivates the thrombin
- and prevents clotting.

→ Heparin also affect the coagulation factor ~~X~~, and ~~XII~~, ~~XI~~, ~~IX~~, ~~VIII~~
- and prevents coagulation.

→ Heparin is used subcutaneously ($\vec{sc}$) or Intravenously ($\vec{IV}$).

⇒ Uses :-

⇝ 1.) Deep Vein Thrombosis :-

→ Thrombosis occurs into the deep veins of the body.
→ It mainly occurs due to long term immobilisation.
→ It occurs due to abnormal clotting into the vessels.

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→ Heparin prevents the clotting into the vessels.

→ Heparin can not produce lysis of thrombus but it can prevent further clotting.

→ Heparin is used $\overrightarrow{SC}$ or $\overrightarrow{IV}$ than patient is shifted on oral coagulent.

2.) Myocardial Infuction :-

→ Heparin is used in ecute myocardial Infuction (MI) for preverition of thrombosis.

3.) Unstable Angina :-

→ Heparin is used to prevent thrombosis in unstable angina.

4) Atrial Febrilisation :-

→ Abnormal rhythm of citria.

→ Heparin is used as precaution.

5.) DIC (Disseminated Intravasculer Coagulation) :-

→ In DIC

- abnormal coagulation occurs in intravesculer compartment.

→ Heparin is used to inhibits that abnormal coagulation.

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~> 6.) Vascular Surgery:-

~> 7.) Other Uses:-

- Bypass surgery
- Prostatic heart valve surgery.

=> ADR:-

-> 1.) Bleeding -> Hematuria
-> GIT bleeding
-> Internal bleeding

-> 2.) Thrombocytopenia.

-> 3.) Hypersensitivity Reaction -> Rashes.

-> 4.) Osteoporosis.

-> 5.) Alopecia.

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Que:- LMWH is preffered over UFH ?
(Unfraxinated Heparin) =

-> Size of LMWH is lower than UFH.
-> LMWH is faster absorbed via subcutaneous route.
-> The action of LMWH is longer.
  • So, once a day used.
-> Side effects of LMWH is lesser.
  • Like thrombocytopenia, osteoporosis, alopecia is lower with LMWH.
-> Monitering with APTT (Activate Partial Thromboplastin Time) is not required for LMWH.
  • LMWH acts on factor Xa
  • It is not affecting thrombin.
  • So, APTT is not affected by LMWH.

SQ

Que:- Oral Coagulation ?

=> Name of oral coagulation:-

-> 1.] Coumerine Derivative :-
  - Warfarin
  - Dieumerol
-> 2.] Rivaroxaban.
-> 3.] Phenindione.

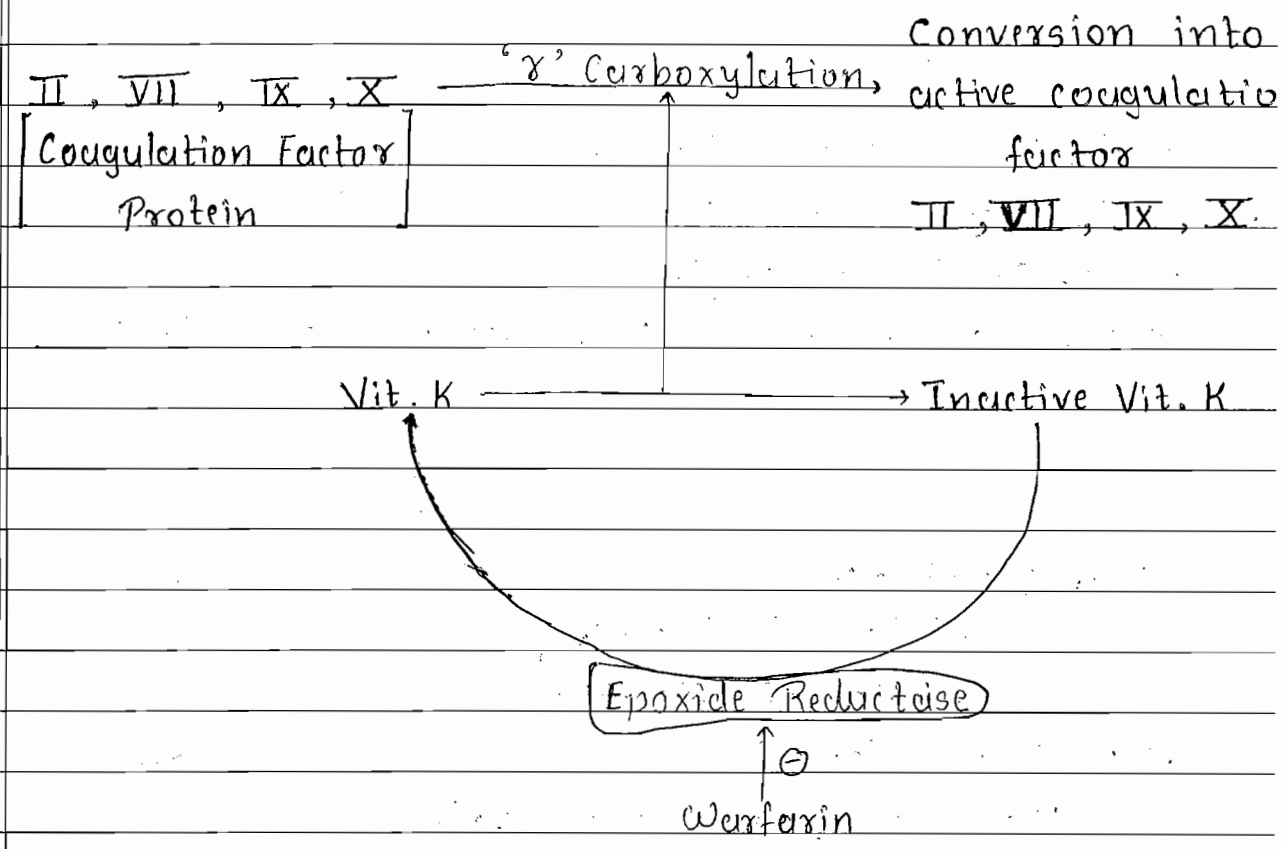
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Que => Description of Warfarin:-

→ It is coumarin derivative.

→ M.O.A.:-

→ Liver:-



→ Coagulation factor II, VII, IX, X are activate by 'γ-Carboxylation'.

→ Vitamin-K is necessary for 'γ-Carboxylation' of this factors.

→ During this reaction vitamin-K is converted into inactive (Vit.K epoxide) form.

→ Warfarin inhibit Epoxide Reductase enzyme and 'γ-Carboxylation' of coagulation factors inhibited.

→ So, it takes time (1-5 days) for action of warfarin.

~> Uses:-

→ i.) Deep vein thrombosis.
→ ii.) Myocardial Infuction.
→ iii.) Unstable angina.
→ iv.) Disseminated Intravascular coagulation.

~> ADR:-

→ 1.] Bleeding:-
- internal bleeding
- External bleeding

→ 2.] Skin Necrosis

→ 3.] Diarrhea ; abdominal pain.

→ 4.] Teratogenic effect
- Warfarin can cross the placenta and affects the featus.
- Hypoplasia of orgens.
- Bleeding tendancy increased.
- Intra uterine death.

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⇝ Contraindication :- (Do not use) :-

→ Bleeding Disorder (Hemophillia);
→ Peptic Ulcer;
→ Liver Damage;
→ Pregnancy;
→ Hemmorage;

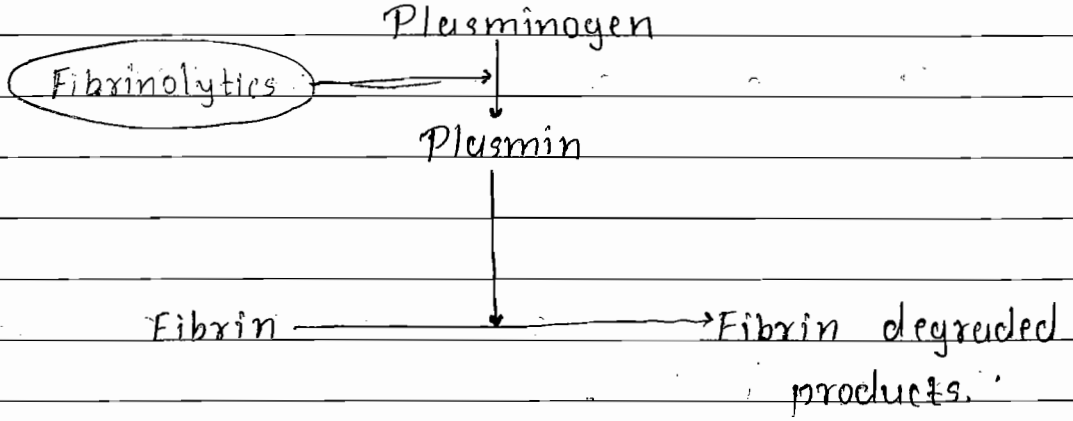
Ques Fibrinolytics :-

→ The drugs which increase the breakdown of fibrin known as fibrinolytics.

⇒ Name of drugs used as fibrinolytics :-

1.) Streptokinase
2.) Urokinase
3.) Alteplase
Reteplase
Tenecteplace.

⇒ M.O.A :-



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→ Fibrinolytics convert plasminogen into plasmin.

→ Plasmin produce degradation of fibrin.

→ So, clot is removed from the circulation

→ Revascularisation of tissue occurs.

⇒ Uses:-

↝ 1.) Myocardial Infaction:-

- Thrombosis inside coronary artery produce, Infaction of myocardium.

- Fibrinolytics is used immidiatly after starting of symptoms.

- Best reasult is possible if they are used within on hour of starting of symptoms.

- They are alternative of PCI.

↝ 2.) Pulmonary Embolism:-

- Fibrinolytic removes the embolus from the circulation.

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3.) Deep Vein Thrombosis:-

→ Fibrinolytics can remove thrombus of deep vein.

⇒ Description of Fibrinolytics:-

1.) Streptokinase:-

→ It's a protein which we get from β - hemolytic streptococci.

→ It is analgenic.

→ Antibodies are produced against it.

→ It can produce anaphylaxis Reaction.

→ So, it should be used only for 1 time

ADR:-
- Bleeding chances are more.
- Hypotension.

2.) Urokinase:-

→ Enzymetic protein.

→ It is obtained from human fetal kidney cell culture.

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→ It is not antigenic.

→ It is not producing hypersensitivity reaction.

~ ADR:-
- bleeding.
- hypertension.

~ 3.) Alteplase-

→ It is obtained by R-DNA technology.

→ Alteplace is more selective plasminogen activator.

→ Non-antigenic.

→ Costly

~ ADR:-
- Less chances of bleeding.

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Ques:- Antiplatelate Drugs:-

→ The drugs which inhibit aggregation of platelates known as antiplatelate drugs.

⇒ Classification:-

→ 1.] Thromboxane A2 Inhibitor:-
- Aspirin (Low dose 50 - 325 mg).

→ 2.] Phosphodiesterase Inhibitor:-
- Dipyridamole.

→ 3.] Purinergic (P_2Y_{12}) receptor antagonist:-
- Ticlopidine.
- Clopidogrel.
- Prasugrel.

→ 4.) GP IIb / IIIa receptor antagonist:-
- Abxicimab.
- Ebtifibatide.

Ques:- Aspirin (as antiplatelate drug) ?

⇒ M.O.A.:-

→ Aspirin is COX enzyme inhibitor.

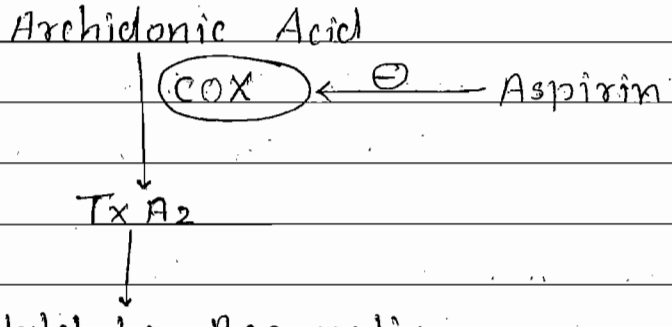
→ So, it reduce the concentration of thromboxane A

→ Thromboxane A2 is responsible for platelate aggregation.

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→ Thromboxane A_2 is responsible for platelate aggregation.

→ Aspirin reduce Tx A_2 concentration so, it works as antiplatelate drug.



=> At higher dose of aspirin :-

→ Aspirin inhibit PgI_2 synthesis.

→ PgI_2 is inhibiting platelate aggregation.
· So, at higher dose aspirin's antiplatelate aggregation action is lost.

Ques- Phosphodiesterase Inhibitor :- (as anti platelate) :-

→ Dipyridamole

=> M.O.A. :-

→ Dipyridamole inhibite phosphodiesterase enzyme.

→ So, cAMP metabolism is reduced.

→ So, cAMP concentration increase.

→ cAMP inhibits platelate aggregation

→ So, Dipyridomole is used as antiplatelate drug.

Phosphodiesterase enzyme ← ⊖ — Dipyridamole

cAMP ——→ degradation

↓

Inhibite platelate aggregation.

Ques. Purinergic receptor antagonist (as antiplatelate) :-

=> Name of Drugs :-

- Ticlopidine.
- Clopidogrel.
- Prasugrel.

=> M.O.A. :-

→ These drugs inhibit the action of ADP.

→ ADP (adinosine Diphosphate) is increasing platelate aggregation.

→ These drugs inhibit ADP release and action So, they inhibite platelate aggregation.

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ADP ⟶ Platelate aggregation

⊖

Purinergic receptor blocker

Ques GP IIb / IIIa receptor antagonist (as antiplatelate)

=> Name of Drugs :-

- Abxicimab
- Eptifibatide.

=> M.O.A. :-

→ GP IIb / IIIa receptors are present on platelates.

→ They take part into final step of aggregation of platelate.

→ GP IIb / IIIa antagonist inhibit this receptor.

→ and prevents platelate aggregation.

→ These are more selective and potent drugs.

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=> Uses:- (Same in above three questions) :-

~> i.) Myocardial Infuction:-

-> Aspirin can be given at the initial stage of myocardial infuction.

-> Aspirin + Clopidogrel is used for long time for prevention of myocardial infuction.

~> ii.) Accute Corronary Syndrome:-

-> Antiplatelate drug is used as treatment and prevention of acute coronary syndrome.

~> iii.) Angina Pectoris:-

-> Antiplatelate drugs used for the prevention & treatment of angina pectoris.

~> iv.) Peripheral Vascular Diseases:-

~> v.) Prostatic Heart Valve Surgery:-

-> after surgery to prevent thrombosis on the valve.

~> vi.) Transient Ischamic Attack:-

-> Antiplatelate drugs are used for the prevention & treatment of TIA.

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=> ADR:-

→ Bleeding

→ Aspirin → Contreindicated in preynancy
→ Rey's Syndrome.

→ Dipyridome → Diarrhea.

Blue:- Iron Therapy:-

→ Oral Preprations:-

- Ferrous Gluconate
- Ferrous fumerate
- Ferrous Sulfate
- Colloidle ferric hydroxide
- Carbonyl iron.

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→ Parentral Preprations:-

- Iron Sorbitol
- Iron dextran
- Ferrous Carboxymertol.

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Ques- Iron Therapy:-

→ Oral iron is used for the iron deficiency.

⇒ Uses:-
→ Iron deficiency anemia.
→ Pregnancy.
→ menstrual bleeding.

⇒ In pregnancy:-

→ Iron (100mg) along with Folic acid (0.5 mg) used.
→ During 2nd trimister of pregnancy for around 100 days.

⇒ In Iron defeciency anemia:-
- Iron around 200 mg is used.
- It is given along with vitamin C because vitamin C increase the absorption of oral iron.

Duration:- In 4 - 8 weeks haemoglobin count is corrected.
- Then therapy should be contineoud for almost 3 - 6 months to replanish iron store.

⇒ Indication for parentral iron therapy:-
- Severe anemia
- Intolerent (vomiting, nausea, GIT disturbances) to oral therapy
- Poor pregnancy of oral therapy.

- Malabsorption of iron.

⇝ ADR :-
- Metalic Test
- Nausea, Vomiting.

→ Dose of iron in parentral therapy is calculated by following equation.

Dose of parental Iron (In mg)	= 4.4 ×	Body Weight of Person	×	Haemoglobin Deficient.

⇒ Intramuscularly :-

→ Parentral iron can be administered deep intramuscular injection.
→ It is done by 'z' technique.
→ Iron solution do staining of the skin.
→ By 'z' technique, solution which is administered at the site, do not come back and staining is prevented.
→ Daily recommended dose in adult is 100 mg/day.

⇒ Intravenous Therapy :-

→ Calculated dose is diluted in 500 mg of saline.
→ It is complete within 6-8 hrs.

-> Infection should be stopped.
- If person starts developing brethlessne chest discomfort.

-> For iron toxicity antidote is -> desteroxan

~> ADR :-
- Painful injection.
- Abscess.
- Skin -> staining.
- Nausia ; Vomitting.

Que. Skeletal Muscle Relaxant :-

-> Skeletal muscle Relaxant.

=> 1.) Centrally acting :-
-> Benzodizapam
- Dizopam
- Lorezapam
- Midazolam
-> Methacarbamol
-> Gobapentine

=> 2.) Peripheral acting :-

~> i.) Depolarising :- (Non - compititive) :-
-> Succinyl choline
-> Decamethonium.

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ans ii) Competitive :- (Non-depending) :-

-> DTC (Tubex curonium).
-> Vacuronium.
-> Rocuronium.
-> Pancuronium.
-> Atracurium.
-> Mivacurium.

ans iii) Directly acting on Skeletal Muscle :-

-> Dentroline.

Que- Succinyl choline as skeletal muscle relaxant?

-> Succinyl choline is ~~a depolarising~~, a depolerising.
-> skeletal muscle relaxant.

-> Non-competitive action.

=> M.O.A. :-

-> It acts at neuromuscular junction.

-> It depolarise the skeletal muscle.

-> So, muscle do not respond to Ach.

-> So, it leads to flaccid paralysis of muscle.

-> And muscle relaxation occurs.

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-> It is metabolised by pseudocholinesterase.

-> It's action is short - lasting (3 - 8 mins).

=> Uses:-

-> 1.) It is used along with general anesthesia for producing good muscle relaxation. (during surgery ; Minor orthopedic procedure ; endoscopic surgery).

-> 2.) Short surgical procedure.

-> 3.) In status epilepticus for reducing traumas.

=> ADR:-

-> Apnea (Stoppage of respiration)
-> Muscle pain.
-> Vagal stimulation -> bradycardia.

Ques- Hypolipidemic Drugs:-

-> The drugs which lower plasma ~~hypolipidemic~~ cholesterol levels.

-> Classification is given below:-

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=> Classification :-

~> i.) HMG CoA Reductase Inhibitors :-

- Atorvastatin
- Rosuvastatin
- Lovastatin
- Simvastatin

~> ii.) Bile acid binding Resins :-

- Cholestyramine

~> iii.) Dietry Cholesterol absorption inhibitors :-

- Ezetimide.

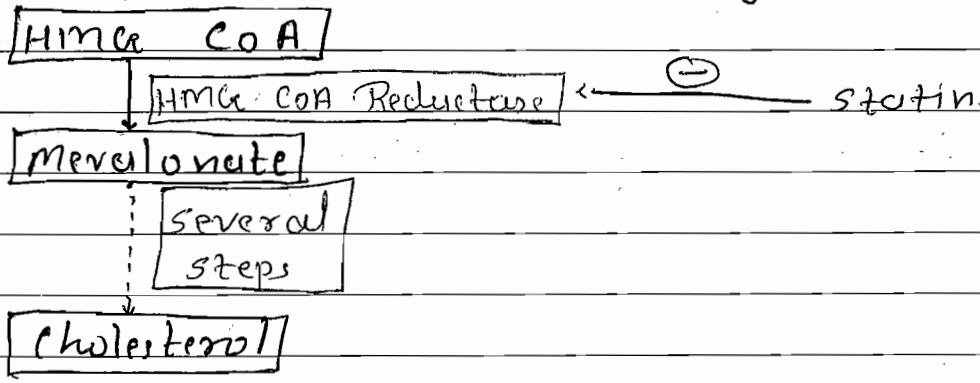
~> iv.) Nicotinic acid :-

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~> HMG CoA Reductase Inhibitors :-

=> M.O.A. :-

- It prevents cholesterol synthesis.



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→ So, LDL and VLDL production is also reduced.

⇒ Uses:-

→ It is used in hyperlipidemia

→ Post Myocardial Infaction for prevention of athrosclerosis.

→ In Diabetes Mellitus for prevention of hyperlipidemia.

⇒ ADR:-

→ Hepatotoxicity.
→ Nausea, Vomiting, Diarrhea
→ Headache.

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Que:- Treatment therapy for myocardial Infuction?

→ Pharmacotherapy ~~for~~ of myocardial Infuction.
→ Death of the cells of myocardial known as myocardial Infuction which most commonly occurs due to reduce blood supply.
- Most commonly because of thrombosis of cardiac vessels.

⇒ Clinicle Features:-

→ Severe chest pain
→ Pain is radiating towards ulnar border of upperlimb and Jaw.

⇒ Diagnosis:-
- ECG
- Cardiac Markers.

⇒ Treatment:-
→ Aspirin at low dose. (60 - 325 mg) (antiplatelate drug) (Tablet)
- If the patient having allergy for aspirin then clopidogrel is used.
- O_2 therapy. (2 - 4 L/min)
- For pain relief → Opioids - Morphine (10 mg)
(Intravenous) Or Pethidine (50 - 75 mg)
- For removal of thrombus
· Thrombolytic Therapy → streptokinase, Urokinase; Altiplase.
Or
PCI (Percutaneus Corronary Intervension).

- Parentral Anticoagulant - Heparin

Or

LMWH (Low Molecular weight Heparin)

- β - blockers → It is used if the patient is not hypotensive.
  → But as a regular therapy it is used because it is cardioprotective.
- Nitrates → Intravenous Nitrates to reduce pain and cardiac workload.
- ACE Inhibitors → It is used for prevention of cardiac remodeling.
- Acidocic :- Sodium bicarbonate should be given.
- Stating :- For reducing cholesterol level in blood.

- ~~#~~

Ans After termination of acute attack drug therapy used are :-

- Antiplatelate Drug
- Statin
- β - blocker / ACE inhibitor.

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Ques- Treatment therapy for Congestive Heart Feillure?

→ When the heart is not able to pump enough amount of blood known as heart feillure.

→ It leads to accumulation of blood within the heart known as congestive heart feillure. (CHF).

CHF

- Back Pressure
  - Pulmonary Oedema; Breathlessness
  - Preload ↑
- Blood to Tissue is decreased
  - (Kidney)
  - Renin Angiotensine aldosterone System is stimulated
    - → Vasoconstriction.
    - → Na^+, Water retension
      - → Oedema
- Forward Pressure
  - After load ↑

⇒ **Treatment Goal :-**

↝ **Reduce Preload :-**

→ **Diuretics :-**
- Loop diuretics → Furosemide.
- Thiazides → Hydroclorthiazide.
- K^+ sparing diuretics → Spironolectone.

→ **Arterial and Venodilators :-**

Both
- ACE Inhibitors → Enalapril ; Captopril ; Ramipril
- ARB → Losartan; Candesartan ; Valsartan.

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- Nitrates :-
  - GTN
  - Isosorbide dinitrate
  - Isosorbide mononitrate.

(Veno dilator)

(Arterio dilator) + Hydralazine.

⇒ For increase force of Contraction :-

→ Sympathomimetics :-
  - Dopamin
  - Dobutamin

→ Cardiac Glycocytes :-
  - Digoxine.

⇒ For prevention of cardiac arrythmia & Overstimulatio of heart :-

→ β - blockers
  - Atenolol
  - Metaprolol
  - Bisoprolol

⇒ CHF is classified into different stages :-
i.) Mild CHF
ii.) Moderate CHF
iii.) Severe CHF
iv.) Refractory CHF.

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Que:- Cardiac Glycosides ?

ans Name of Drugs:-
- Digoxin
- Digitoxin.

ans Mechanism of Action:-

- Glycosides inhibit $Na^+ - K^+$ ATPase pump.
- Na^+ Concentration is increased intracellularly.
- Ca^{+2} is increased intracellylarly.
- Cardiac muscle contractibility increase.
- Force of contraction is increased.
- Cardiac output is increased.
- But heart rate is not increased.

ans Uses:-

1.] CHF:-
- In this condition heart is not able to pump enough amount of blood.
- Cardiac Glycosides increase force of contraction & increase cardiac output.
- So, CHF is treated.

2.] Cardiac Arrythmia:-
- Cardiac glycolysides reduce the impulse production & increase PR interval.
- So, they can be used to treat arrythmia.

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ADR I-

- Nausia , Vomiting ;
- Head ache ;
- Gynecomastia ;
- Restlessness ;
- Body accumulation ;
- AV block .

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★CENTRAL NERVEOUS SYSTEM★

Que 1- Sedatives & Hypnotics ?

⇒ Sedatives :- Substances which reduce the excitement and calm the patient.

⇒ Hypnotics :- Substances which induce the sleep.

Introduction :-

Physiology of Sleep :-

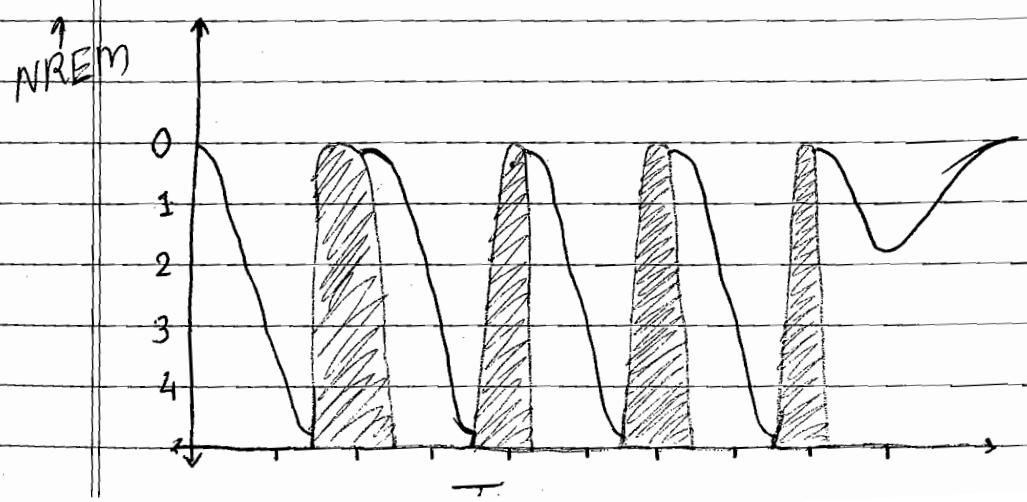
There are two types of sleep.

1.] NREM sleep (Non Rapid Eyeball movement) :-

- Stages :- 0, 1, 2, 3, 4.

2.] REM sleep (Rapid Eye ball Movement) :-

- Rapid Eyeball movement sleep.
- Dreems occurs during this sleep.
- 20% - 30% of sleep.



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-> During sleep there is a cycle of (NREM - REM) sleep.

-> Person spends majority of the time in stage 2 ; NREM movement.

-> Perticular system of brainstem is responsible for conciousness and awakening.

-> Sedative and ~~hypnotics~~ hypnotics after reticular system.

Que :- Classification of Sedatives and Hypnotics ?

ans 1.) Barbiturates :-

i.) Long acting :-
- Phenobarbitone

ii.) Short acting :-
- Butobarbitone

iii.) Ultra short acting :-
- Thiopentone Sodium.

ans 2.) Benzodizepines :-

Hypnotics	Antianxiety :-	Anticonvulsant
• Diazepam	• Diazepam	• Diazepam
• Alprazolam	• Alprazolam	• Lorazepam
• Nitrazepam	• Lorazepam	• Clonazepam.

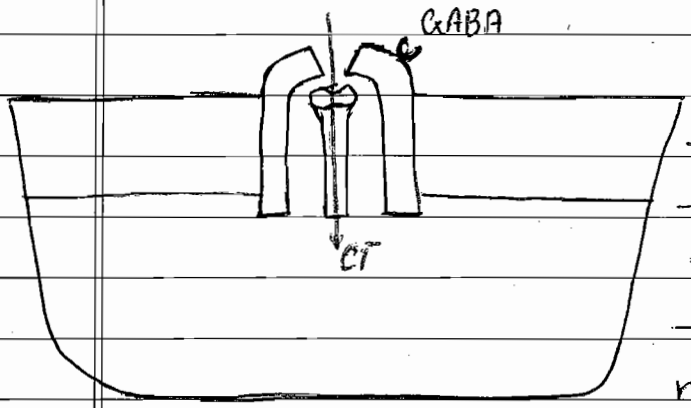
ans 3.) Non - Benzodizepine hypnotics :-
- Zolpidem.
- Zopiclone.
- Zaleplone.

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Ques- M.O.A. ; Pharmacological action ; Therapeutic action and ADR of Diazepam (Benzodiazepam)?

ans M.O.A.:-

→ Benzodiazepanes binds on the benzodiazepam receptors of GABA channels.



→ and open the cloride channel.
→ Cloride goes inside neurone
→ It produce inhibition of neuron
→ It affects reticular system, midbrain and Lymbic system.

(DIAZEPAM AC) (Pharmacological action)

Diagnosis & Minoroperation
Insomia
A
Z
E
Precanesthatic Medication
Anxiety
Muscle Relaxant
Alcohol withdrawl
Consious seduction.

=> Pharmacological action & Therapeutic Uses :-

1.) Insomia :-

- Difficulty in initiation of sleep known as Insomia.

Types :-

- transient Insomia → < 3 days.
- short Term Insomia → 3 days - 3 weeks.
- Long Term Insomia → > 3 weeks.

- Benzodizepines are preffered (used) for insomia.
- They are better than barbiturates because
  - = Dipendance is less.
  - Tolerance is less
  - morning drowsiness and headache is less.
  - Sleep pattern is not disturbed.
  - benzodizepines affects (increase). stage II NREM sleep.
  - Benzodizepanes do not affect REM pattern.
- Benzodizepanes should not be used for long time. Otherwise they also produce dependance and tolerance.

2.) Diagnosis and Minor operation :-

- Diagnosis can be used in minor surgery because it produce Amnesia ; Anxiety effects also.

3.) Preanesthetic Medication :-

- Diazepam reduce the anxiety and calm the patient
- So, used as preanesthetic medication.

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→ 4.) Anxiety :-
- Diazepam also affects neurons of limbic system and reduce the anxiety.

→ 5.) Muscular Relaxant :-
- Dizepam is centrally acting skeletal muscle relaxant.

→ 6.) Alcohol Withdrawal :-
- Diazepam can be used to relieves the symptoms of alchohol withdrawal.

→ 7.) Anticonvulsant :-
- Diezepam reduce the impulse sensation.
- So, it can be used to treat convulsion.
- ex. febric convulsion status epilepticus.

→ 8.) Concious Sedation :-
- For unco-operative patient, diazepam can be used to calm the patient.

⇒ ADR :-
- Dependance
- Tolerance
- Drowsiness → In the morning (after sleep)
- Headache
- Amnesia
- Hypotension (BP↓).
- Respiratory depresion

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=> Diazepam Toxicity :-

Antidote → Flumazenil (IV)
(antagonist Benzodiazepine Receptor).

Ques Barbiturates :-

ans Name of barbiturates :-
- Phenobarbitone.
- Butobarbitone.
- Thiopentone Sodium.

ans M.O.A. :-
- Barbiturates binds with GABA channel on [barbiturate receptors] α / β.
- Opens cloride channel.
- Cloride goes inside the neuron.
- Hyperpolarisation of neuron.
- Neuron → inhibited.

ans Pharmacological action & Uses :-

→ 1.) Insomnia :-
- Barbiturates can be used.
- Barbiturates affects normal pattern of sleep.
- They affect REM, NREM stage II, IV sleep.
- Safety margin is less.
- Drowsiness, headache, chances are less.

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→ 2.) Induction of General Anesthesia:-

- Thiopentone sodium is used for induction of General Anesthesia.
- It is faster acting and act for very short duration.

→ 3.) Anticonvulsant:-

- Barbiturates blocks impulse generation in neurons.
- So; they can be given as anticonvulsant.

→ 4.) Enzyme Induction to Jaundice:-

- Phenobarbitone induce enzymes of billirubin conjugation.
- So; it can be used for jaundice in child.

→ ADR:-

- Dependance
- Tolerance
- Drowsiness.
- Headache.
- Respiratory depression.
- Rebound increase in sleep (REM)
- Hypotension.
- Anemia.

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Q:- Nonbenzodiazepines Hypnotics :-

Ans Name of Drugs :-

- Zolpidem
- Zaleplone.
- Zopiclone.

Ans M.O.A. :-

- These drugs binds with GABA channel. (other than benzodiazepam receptors).
- Open cloride channels.
- Cloride goes inside the neurone.
- Hypopolerisation of neurone occurs.
- Neuronal inhibition.

Ans Pharmacological action and use :-

Insomnia :-

- These drugs are good for insomnia.
- Because
  - Short acting;
  - Morning drowsiness ; headache less;
  - Dependance less ;
  - Tolerance is less ;
- These drugs are less anticonvulsa and muscle relaxant.

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ans ADR:-
- Nausia
- Vomiting
- Headache
- Drowsiness.

Que- Drugs used in Parkinson's Diesease ?

ans 1.] Drugs affecting Dopaminergic System:-

[i.] Dopaminergic Precursor:-
- Levodopa

[ii.] Peripheral Decarboxylase inhibitor:-
- Carbidopa

[iii.] COMT Inhibitor:-
- Entucarpone.
- Tolcarpone.

[iv.] MAO Inhibitor:-(Mono Amino Oxidase):-
- Selegiline.

[v.] Dopamin Agonist:-
- Bromocriptine
- Ropinirole
- Pramipexole

[vi.] Dopamin Facilitator:-
- Amantidine.

2.] Drug affecting Cholinergic System :-

[i.] Centrally acting anticholinergic :-
- Benzhexol
- Biperidine
- Procyclidine

[ii.] Antihistaminic :-
- Promethazine
- Orphanedrine.

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Que :- M.O.A. ; Pharmacological action ; and ADR of Levodopa

→ It is a dopamin precursor drug.
→ As dopamine can not cross Blood Brain Barrier ;
Dopamine is given in the form of Levodopa.

M.O.A. :-

→ Levodopa crosses the Blood Brain Barrier and reach to the brain tissue.
→ Levodopa is converted into dopamine.
→ Dopamine acts on D_2 receptor of Basal Ganglia.
- Because it is D_2 receptor agonist.

→ And inhibits the neurons.
→ So, symptoms of parkinson's diesease is resolved.
→ Rigidity, Tremor, Hypokinesia is resolved.

Ans Uses:-

→ Levodopa is used to treat parkinson's diseseuse.

→ Levodopa is combined with carbidopa. (Why?) So

- Levodopa is decarboxylated into peripheral tissue.
- So, very less amount is reaching into the brain.
- and peripheral side effects are increase.
- Carbidopa cannot cross Blood Brain Barrier.
- It is acting on peripheral decarboxylase enzyme.
- So, peripheral conversion of levodopa is prevented.
- And high amount of levodopa reach into the brain.

Ans ADR:-

→ It produce nausea and vomiting by stimulating CTZ (Cemo Trigger Zone) of brainstem.
→ Hypotension
→ Tachycardia; Arrythmia
→ Hallutination; irrelevant thinking psycosis like condition.
→ Confusion; Headache.
→ Behaviour Changes.
→ Ataxia;

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-> On and Off phenomena;

- Drugs effect goes ——> and Symptoms starts 'Off'

- Drug is taken ——> Symptoms disappear. 'On'

- Due to long term treatment with Levodopa - On - off phenomena occurs.
- Levodopa is combined with other drugs for prevention of 'On - off' effects.

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Que:- Carbamazepine ?

→ Carbamazepine is from iminostilbene group.

⇒ M.O.A.:-

→ Carbamazepine increase duration of inactivated sodium channel or prolongation of inactivated sodium channel.
→ So, that neuronal stimulation is reduced and epilepsy treated.

→ It is bound to protein
→ It is having ADH like action.
→ It can reduce urine volume and produce water intoxication.
→ It is having autoinduction effect.
[It increase it's own metabolism].

⇒ Uses:-
→ i.] GTCS (Generalise Tonic Clonic Seizure)
→ ii.] SPS (Simple Partiele Seizure)
→ iii.) CPS (Complex Partiele Seizure)
→ iv.) Trigeminal Neuralgia (Pain along the distribution of trigeminal nerve).
→ v.) Acute Mania.
→ vi.) Bipolar Disorder.

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⇒ ADR:-
- Drowsiness.
- Sedation.
- Vertigo.
- Urinery Retention.
- Hypersensitivity reaction.
- Hepatitis.
- Bonemarrow suppresion.

Que- Valproic Acid or Sodium Valproate ?

→ Broad Spectrum anti-epileptic drug.

⇒ M.O.A.:-

→ It do prolongation of inactivated sodium channel.
→ It inhibits calcium current in the neurons.
→ It activates GABA channel.
- By increasing GABA production and decreasing GABA production.
- So, Cl^- goes inside.
- Hyperpolarisation of neuron occur.
→ Thus, sodium valporate inhibit neuron stimulation by 3 different channels.
→ It is plasma protein bound.
→ It is metabolise into liver and excreated through urine.

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⇒ Uses:-

→ i.] First line drug in
- Abscene Seizure.
- Myoclonic Seizure.
- Atonic Seizure.

→ ii.] Second line drug in
- GTCS (Generalise Tonic Clonic Seizure)
- SPS (Simple Partiele Seizure)
- CPS (Complex Partiele Seizure)

→ iii.] Acute Manics.
→ iv.] Bipolar Disorder.
→ v.] Prophylaxis of Migraine.

⇒ ADR:- (VALPROATE).

→ Vomiting
→ Anorexics
→ Liver Damage
- It is not given in children less than 3 years of age.
- because it produce fulminent hepatitis in children.
→ Elevated liver enzyme
→ Pancreatitis
→ Rashes
→ Alopecia (hair loss)
→ Teratogenic effect.
- Neural tube defect (spinabifida).
→ Oral - facial dysmorphism

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Ques- Status Epilepticus ?

-> When epilepsy attack continue > 30 minutes or person having two or more than two attack without gaining conciousness in between.

=> Treatment :-

-> Hospitalise the patient.
-> Maintain air coay.
-> Administer oxygen.
-> Blood sample for testing
-> Water and electrolyte balance (Glucose, Electrolyt

ans Step - I :-

-> Lorezepam 0.1 mg/kg $\vec{IV}$

Or

Diazepam 10 mg/kg $\vec{IV}$

Repeate after 10 min if necessory

Or

fosphenitone 20 mg/kg $\vec{IV}$

Or

Pheniton 20mg/kg $\vec{IV}$

- If seizure contineus.

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ans Step - II :-

→ Phenobarbitone 10 - 15 mg / kg IV

→ If seizure continues.

ans Step - III :-

→ General anesthesia
i.v. Midazolam
or
i.v. Propofol
- BP mesurment continews
- Maintain Ventilation.

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Que- Epilepsy :-

=> Introduction :-

→ PCD :- Paroxysmal Cerebral (Sudden brain)
Dysrythmia
(abnormal impulse generation).

→ Epilepsy is a Neurological disorder in which sudden uncontrolable abnormal impulse generation occurs into the brain

→ It can be sensory and motor.

→ motor epilepsy also known as convulsion.

Types

Generalised Seizure	Partial Seizure
- GTCS :- Generalised Tonic Clonic Seizure	- Simple Partial Seizure
- There is general involvment of whole body muscle	- Group muscle involved.
- Absence Seizure :- Mostly child affected visible become fixed loss of conciousness Amnesia.	- Complete Partial Seizu Group of muscle involv + other symptoms (loss of conciousness mental disterbance).
- Atonic Seizure :- Muscle tone is lost.	

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- Myoclonic Seizure:- Shock like momentary construction of muscle

- Simple Partial Seizure:- may become generalised.

Que:- Classification of drugs used in epilepsy or Antiepileptic drugs?

-> Phenytoin
- Fosphenytoin
- Carbamazepine
- Oxcarbamazepine
- Valproic acid (Sodium Valproate)

-> Barbiturates:-
- Phenobarbitone

-> Benzodiazepines:-
- Diazepam
- Lorazepam
- Clonazepam

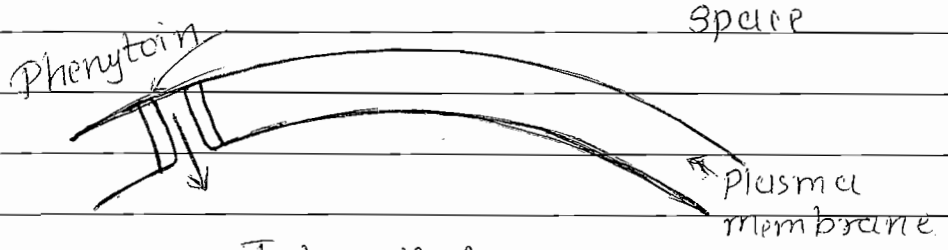
-> GABA Pentine:-
- Etho succinide

or Newer Antiepileptics:-
- Levetiracetam
- Topiramate
- Vigabetrine

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→ Phenytoin :-

=> M.O.A. :-



→ Phenytoin binds with Na^+ channel.
→ and it prolongs the inactivation of Na^+ channel
→ So, that Na^+ channels do not activated.
→ So, impulse generation is reduced.
→ So, epilepsy is treated.

=> Use :-

1.) GTCS (Generalize Tonic Clonic Seizure)
2.) Partial Seizure
3.) Status-epilepticus - Continuous attack of Seizure > 30 min.
4.) Trigeminal Neuralgia

→ Pain attack on the face because of abnormal stimulation to trigeminal nerve.
→ Phenytoin can relive the attack of trigeminal Neuralgia.

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⇒ ADR :-

→ Hirsuitism :- Abnormal hair growth on face on female.

→ Hypertrophy of gum.

→ Hypersensitivity reaction.

→ Hypoplastic bone marrow - megaloblastic anemia

→ Hepatic damage

→ Pregnancy fetal hydantoin syndrome
- cleft pulate
- Microcephaly
- Hare lip.

⇒ CNS :-
M - Mental Confusion
D - Drowsiness
B - Behaviour Disturbance
H - Hallucination
A - Ataxia
V - Vertigo

⇒ CVS :-
Hypotension
Cardiac arrythmia.

⇒ GIT :-
- Nausia
- Vomitting

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☆ Opioids ☆

☆ It is an alkaloids which we get from a plant Papaver Somniferum.

- Morphine is natural.
- We have different type of opioids analgesic.

Que:- Classification of Opioids?

⇒ Natural:-
- Morphine
- Codeine
- Papeverine
- Noscapine

⇒ Semi Synthetic:-
- Pholcodeine
- Hydromorphine

⇒ Synthetic:-
- Pathidine
- Tramadole
- Fentenyl
- Alphentenyl
- Remiphentanyl

⇒ Opioid Agonist:-
- Pentazocine.

⇒ μ-receptor agonist, κ-receptor antagonist:-
- Buprenorphine.

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⇒ Pharmacological Action:-

i.) CNS:-

→ Morphine acts on μ, κ, δ receptor located at spinal and supraspinal region.
→ It modify the pain conduction.
→ Morphine stimulates analgesic pathway so; that pain pathway is inhibited.
→ So, Subjective feeling of pain is reduced.
→ Anxiety and fear is reduced.
→ Person become calm.
→ It also produce euphoric effect.
→ Reduce conciousness.

ii.) Eye:-

→ Morphine acts on midbrain on 3rd cranial nerve nucleous and stimulate it.
→ Produce miosis.

iii.) Respiratory System:-

→ Morphine supress respiratory centre and produce reduction in respiratory rate.
→ Increase CO_2 accumulation.

iv.) Physicle & Psycological:-

→ Morphine reduce physicle and Psycological dependance.

v.) CVS:-

→ Morphine increase histamine release and produce peripheral vasodilatation.
→ Vegal stimulation.
→ Hypotension is produced.

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ans vi.) GIT :-

→ Morphine stimulate GIT sphincters and close them.

→ Mostly anal sphincter and pyloric sphincter are closed.

→ So, it produce constipation.

ans vii.) Urinary Bladder :-

→ Morphine increase tone of bladder sphincter

→ Closing of bladder sphincter.

→ Urinary Retention.

ans viii.) Cough :-

→ Morphine inhibits cough centre and supress cough.

ans ix.) Hypothermia :-

→ Morphine reduce body temprature.

ans x.) CTZ :-

→ Morphine stimulates vomiting centre and produce Nausia and vomiting.

ans xi.) Skin :-

→ ~~Morphine stimulates increase reduce and reduce~~

→ Morphine stimulates Histamin releuse in skin and produce itching.

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xii.) Billiary Tract :-

→ Morphine stimulates sphincter of a urinary tract and close it.

⇒ Uses :-

i.) Analgesic :-

→ Opioids are prefered drug for releving pain.

→ Their efficacy for reducing pain is higher than NSAIDs.

→ ex. → Myocardial Infarction.
→ Burns
→ Bullet Injuries
→ Pain due to accidental injuries
→ Cancer
→ Renal Stone Pain.

ii.) Preanasthetic Medication :-

→ Opioids reduce Anxiety and calm the person.

→ Minor operations can also be performed after giving opioids.

iii.) Diarrhea :-

→ Opioids produce symptomatic relief in diarrhea by reducing frequency of passing stool.

iv.) [redacted] Cough :-

→ Opioids supress cough centre and reduce production of cough.

→ So, they are used in dry cough.

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v.) Pulmonary Oedema:-

-> Opioids produce peripheral vasodilatation.
-> So, blood is distributed towards peripheral parts of the body.
-> So, fluid accumulation in lung is reduced.
-> Symptomatic Relief occurs.

=> [redacted]

=> ADR:-

-> Difficulty in vision.
-> Respiratory depression.
-> Physicle & Psycologicle dependance.
-> Hypotension
-> Itching
-> Nausia & Vomiting
-> Sedation
-> Euphoria
-> Constipation.
-> Difficulty in micturation.

=> Contraindication :-

-> Bronchi asthma
-> Head injury -> because morphine increase intracranial tension by decrease CO_2 expiration.
-> Hypotension
-> Infants -> Morphine produce severe respiratory depression in infants.

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Que:- Difference between Morphine & Pathidine ?

Morphine	Pathidine
-> It is natural opioid.	-> It is synthetic opioid.
-> Analgesic dose -> 10 mg 10th time more potent than pathidine	-> Analgesic dose -> 100mg 1/10th time potent dose than morphine.
-> Constipation, Urinary Retention, Miosis. -> more with Morphine	-> Constipation, Urinary Retention, Miosis -> Less with Pathidine.
-> Having cough suppresent effect	-> It is not having cough supresent effect.
-> Produce severe Respiratory depression	-> Pathidine produce less respiratory depression.
-> Rapid action but long duration of action	-> Rapid action and short duration of action.

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Que:- Morphine Poisioning ?

→ When morphine is consumed in higher dose, produce poisioning.

ans Clinicle Features are as beloco :-

- Miosis (Pin Point Pupil).
- Sedation; Drowsiness
- Respiratory Depression
- Hypotension
- Nausea ; Vomiting
- Constipation ; Urine Retension
- Coma
- Death (May be due to Respiratory Failure).

ans Treatment:-

→ Stop further intake of morphine
→ Hospitalisation of patient.
→ Airway Support
  • Start ventilation
→ Maintain Circulation
  • Fluid Therapy
→ Gastric Lavage
  • Potassium Permangenate ($KMNO_4$) is used for gastric lavage.
→ Naloxone (Opioid antagonist). (0.4 - 0.8 mg $\vec{IV}$)
→ Stop Naloxone after the Respiratory become normal
→ Maintain Vital function.

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Que:- Classification of Antipsycotic Drugs ?

ans Typical :-
- Chlorpromazine
- Triflupromazine
- Haloperidole
- Trifluperidole
- Proclorperazine

ans Atypical Antipsycotic :-
- Risperidone
- Olanzapine
- Clozapine
- Ziprasidone
- Aripiprazole

Que:- M.O.A, Pharmacological action; Uses and ADR of Clopromazine (CPZ) ?

ans M.O.A. :-
-> Chlopromazine blocks D_2 receptors of lymbic system and other cortical areas.
-> So, Antipsycotic effect is produced.

=> Other Actions of Chlopramazine :-
-> Block CTZ -> stop Nausia, Vomiting.
-> Block α_1 -> Postual Hypotension.
-> Block D_2 receptor -> EPS (Extra Pyramidal Side Effects) of Basal Ganglia (Parkinson's diesease like condition).
-> Block Muscarenic -> Dryness of mouth;
receptor -> Decrease secretion;
-> Constipation

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~> Pharmacological Action:-

=> CNS:-
- Chlorpromazine reduce ; thinking reduce; Hallutination Reduce Insomnia ; Change the Behaviour ; ↓ anxiety.
- By blocking Basal Ganglia produce rigidity and tremor.

=> CVS:-
- Chlorpromazine blocks α_1 receptors and produc hypotension.

=> GIT:-
- CPZ reduce Nausea and Vomiting.
- By Antimuscaranic action → Reduce Secretion ; decrease peristalsis and Constipation.

=> Urinary Bladder:-
- May produce urinary retention

=> Endocrine:-
- CPZ blocks D_2 receptors and increase prolactin level
- In female → Oligomenorrhea.
Or
Amenorrhea.
- In male → Gynacomastia.

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ans Uses:-

=> Mania:- Person having elevated mood; fights of idea; Insomia.
-> CPZ is used for treating Mania.
-> In acute mania; Halloperidol; is used by Intravenous Route.

=> Antiemitic:- CPZ is used to reduce nausea and vomiting.

=> Schizophrenia:- It is psycological disorder in which person is not able to understand difference between reality & dreams.
-> It is characterised by Hallutination, delusion, abnormal social behaviour.
-> CPZ can be used to treat schizophrenia.
-> Now a days Atypicle antipsycotics are used more.

=> Interactive Hiccough:- CPZ can be used for treating Hiccough.

ans ADR:-
-> Perkinson's Diesease:-
- Hypokinesia (↓ Movement)
- Rigidity
- Tremor
-> Abnormal Muscular Movement
-> Hypotension
-> Dryness of mouth.

→ Oligomenorrhea ; Amenorrhea.
→ Gynecomastia
→ Urinary Retention
→ Constipation
→ Weight gain
→ Hypersensitivity reaction
→ Loss of Libido (i.e. Sexual Desire).

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Ques Atypical Antipsycotics ?

⇒ Name of Atypical Antipsycotics :-
- Clozapin
- Olenzepin
- Ziprasidone
- Risperidone
- Ariprazole

⇒ M.O.A. :-
→ These drugs acts by blocking $5HT_2$ receptors of lymbic cortex.
→ So, ~~xxxxxxxxxx~~ produce antipsycotic effect.

So :- What are the benefits of atypical antipsycotics over typical antipsycotics ?
→ Atypical antipsycotics having very weak D_2 blocking action.
→ So, extra pyramidal side effects (Parkinson's Diseases) are not produced.
→ Also block α_1 and muscaranic receptors.

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⇒ Uses:-

↝ i.) Schizophrenia:-

→ Atypical antipsycotics are prefered.

→ They treat positive (Hallutination thoughts) as well as negetive (lack of Motivation; Social withdrawal) symptoms of schizophrenia.

↝ ii.) Mania:-

→ Person having elevated mood; fights of ideas; Insomia.

→ CPZ is used for treating mania.

→ In acute mania; Halloperidol is used by Intravenous route.

⇒ ADR:-

→ Weight Gain;

→ Abnormal Muscular Movement;

→ Sedation;

→ Drowsiness.

Que:- Antidepressants?

→ Depression is a condition associated with 'low mood' or 'sadness'.

⇒ Other Symptoms:-

→ Lack of interest.

→ Sleep disturbance

→ Sucidal Ideas.

→ The drugs used to elevate the person's mood work as antidepressants.

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=> Classification of Drugs:-

ans i.) TCA :- (Tricyclic Antidepressants):-
- Trimipramine
- Imipramine
- Amitriptiline
- Nortriptiline
- Doxepine
- Chlomipramine

ans ii.) SSRI :- (Selective Seratonine Reuptake Inhibitors):-
- Fluoxatine
- Sertraline
- Escitelopram
- Citalopram
- Peroxetine.

ans iii.) SNRI :- (Seratonine Noradrinaline Reuptake Inhibitor):-
- Duloxetine
- Venlafaxine

ans iv.) Atypicle Antidepressant:-
- Trazodone
- Bupropione
- Miansexine

ans v.) MAO - A Inhibitor:-
- Clorgyline.
- Moclobemide.

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Que 1- M.O.A. ; Uses and ADR of TCA ?

=> Drugs:-
- Imipremine
- Trimipremine
- Chlomipremine
- Amitriptiline
- Nortriptiline
- Doxepine

=> M.O.A.:-
-> These drugs inhibit reuptake of Noradrineline (NA) and increase concentration of Noradrineline (NA) into synapse.
-> It stimulate the neuron.
-> Person's mood is elevated.

=> Other Mechanism:-
-> TCA also inhibit reuptake of serotonin.
-> Increase availibility of serotonin at synapse.
-> Person's behaviour and mood changes.
-> TCA also inhibit α receptor and produce postural hypotension.
- Cardiac Arrythmia.
-> TCA also produce anticholinergic action.
- Dryness of mouth.
- Blurring of vision.
- Constipation.
-> Produce sedation.
-> Reduce seizure threshold.

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⇒ Uses :-

↝ i.) Depression :-

- "low mood" state of the person is known as Depression.
- TCAs used in depression.
- It may take few weeks for it's action.
- It increase person's mood and person starts taking interest in surroundings.

↝ ii.) OCD :- (Obsessive Compulsive Disorder) :-

- It is the disorder in which person having an Obsession which is followed by compulsion. (act)
- ex. Checkers, Washers.
- TCA can be used in OCD.
- It prevents the frequency of obsession & compulsion
- and prevents the fluctuation in mood.

↝ iii.) ADHD :- (Attension Deficient Hyperkinetic Disorder)

- Person is hyperkinetic (excessively active) but not able to focus on one thing.
- TCA are also effective in ADHD.

↝ iv.) Noctunal Enuresis :-

- Person pass urine in bed during sleep known as Noctunal Enuresis.
- Psycotherapy (Counselling) is more usefull in Noctunal Enuresis.
- TCA can be used in the treatment of Noctunal Enuresis.

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⇒ v.) Prophylaxis of Migraine :-

→ TCA can be used in the prophylaxis of migraine.

→ It prevents the frequency of attack of migraine.

⇒ vi.) Neuralgic Pain :-

→ Pain because of abnormal stimulation to neurons.

→ TCA can be used to treat such pain.

⇒ ADR :-

→ Postual Hypotension.

→ Cardiac arrythmias.

→ Dryness of mouth.

→ Constipation.

→ Blurring of vision.

→ Sedation.

⇒ Contraindicated in :-

→ Glaucoma.

→ IHD.

→ Epilepsy.

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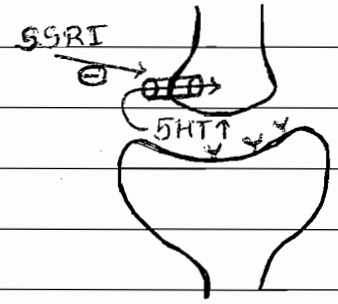
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Que- SSRI (Selective Serotonine Reuptake Inhibitor):-

ine.

ne.

Name of the Drugs:-
- Fluoxentine
- Fluoxamine
- Sertraline
- Citalopram
- Escitalopram
- Paroxetine



M.O.A.:-
→ These drugs inhibit serotonine reuptake.
→ So; Concentration of serotonine increase in the synapse.
→ Serotonine helps for mood elevated.

Benefits of SSRI over TCA:-
→ SSRI do not act on α_1 receptor.
- So, postnual hypotension is not occuring.
→ SSRI do not have anticolinergic activity.
- So; dryness of mouth; constipation; blurring of vision not occuring.
→ SSRI produce very less sedation.
→ SSRI do not produce weight gain.
→ SSRI do not produce convulsion.
→ SSRI better tolerated.
→ Less cardiac toxicity.

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Q) Uses:-

=> i.) Depression:-

→ "Low mood" state of the person is known as Depression.
→ SSRI used in Depression.
→ It may take few weeks for it's action.
→ It increase person's mood and person starts taking interest in surrounding.

=> ii.) Anxiety:-

→ Person having "increased (high) mood".
→ Person having increased conciousness ; Alertness ; increased thoughts.
→ SSRI helps in Reduction of Anxiety.

=> iii.) OCD:- (Obsessive Compulsive Disorder):-

→ It is the disorder in which person having an obsession which is followed by compulsion. (act)
→ ex. Checkers ; Washers.
→ SSRI can be used in OCD.
→ It prevents the frequency of obsession & compulsion.
→ and prevents the fluctuation in mood.

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iv.) Premature ejeculation:-

-> Ejeculation occurs immediately during sexual intercorse.
-> which occurs due to subjective Anxiety.

Or

↑ Sympathetic stimulation.
-> SSRI can be used to treat Premature ejeculation.

ADR:-
-> Insomnia;
-> Headache;
-> GI Disturbances,
- Nausea;
- Vomiting;
-> Loss of libido.

Que:- Bipolar Disorder (Drugs used in bipolar disorder):-

-> Person's mood is changing from mania to Depression and Depression to mania. known as Bipolar Disorder.

-> mood stabiliser used for the treatment of bipolar disorder.

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Name of the Drugs :-

- Carbamazepine } → antiepileptic
- Valproic acid (Sodium Valproate) }
- Olanzepin } → Antipsycotics.
- Risperidone }
- Haloperidol }
- Lithium.

Que:- Lithium :-

⇒ M.O.A. :-

→ Lithium inhibits secondary messenger (IP_3) formation.
→ Lithium mimics the action of Sodium. (acts impluce of Sodium (Na^+))
→ It reduce the release of Dopamine & Noradrenaline.
→ It stabilise the person's mood.
→ So, it is used for Bipolar Disorder.

⇒ Uses :-

i) Bipolar Disorder :-

→ Lithium stabilise the person's mood.
→ So, it prevents mania as well as depression.
→ It takes time for it's action.
→ So, lithium is used as a precaution for bipolar disorder.

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=> ii) Mania :-

-> Lithium can be used for unipolar mania.

ans ADR :-

=> i) Renal :-

-> Decrease ADH Release

- So; increase frequency of urine, (Polyuresi)

↓

Polydypsia (Thirst).

- Known as Diabetes Insipidus.

=> ii) GIT :-

- Nausia
- Vomiting.

=> iii) CNS :-

- Headache
- Confusion
- Ataxia.

=> Lithium overdose may produce

- Cardiac arrythmia
- Coma & Death.

=> Hypothyroidism.

=> Goiter.

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ans Treatment :-

→ Stop the further treatment with lithium
→ Give intravenous Manitol.
→ Give Normal saline to increase Na^+.
→ Hemodyalisis can be done.

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★ Excretory System ★

Que 1- Diuretics ?

-> These are the drugs which increase urine volume.

=> Classification :-

ans 1.] High efficacy diuretics ($Na^+ - K^+ + 2Cl^-$ Inhibitors) :-
- Furosemide
- Torasemide
- Bumetenide

ans 2.] Medium efficacy diuretics :-
i.) Thiazides :-
- Hydrochlorthiazide
- Hydroflumethiazide

ii.) Thiazides like :-
- Chlorthalidone
- Indepemide

ans 3.] Weak diuretics :-
i.) Carbonic anhydrase inhibitor :-
- Acetazolamide

ii.) K^+ sparing diuretics :-
- Spironolectone
- Amiloride

iii.) Osmotic Diuretics :-
- Manitol.
- Glycerol.

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Que:- Furosemide :-

→ It is High efficacy diuretics.

ans M.O.A. :-

→ Furosemide inhibit $Na^+ - K^+ - 2Cl^-$ channel at ascending limb of henle of nephrone.
→ So, Na^+, K^+, Cl^- ions absorption into the blood is reduced.
→ Ion concentration in urine is increased.
→ At DCT ; Na^+ is exchanged by K^+.
→ K^+ excreation is more increased.
→ It is known as Diuresis.

ans Uses :-

ans i.) Oedema :-

→ Accumulation of fluid in the intestinal space (ECF) known as Oedema.
→ Furosemide increase excreation of fluid in urine.
→ Intravesculer fluid is reduced.
→ So, extracellular fluid is shifted into intravesculer compartment.
→ So, accumulated fluid in intestical space is reduced.
→ Oedema is reduced.
→ Furosemide is used in
- Cardiac Oedema
- Cerebral Oedema.
- Pulmonary Oedema
- Hepatic Oedema

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ii.) Hypertension :-

-> Furosemide increase urine formation.
-> Decrease water content of intravascular fluid.
-> So, Cardiac output is reduced.
-> So, Blood pressure is reduced.
-> So, Flurosemide can be used in a treatment of hypertension.
-> It is not used as longterm therapy of hypertension.

Hypercalcemia :-

-> Furosemide increase calcium excretion in urine.
-> It is used with normal saline.
-> So, furosemide is used in Hypercalcemia.

iv.) Fluid Overload :-

-> Furosemide increase fluid excretion in to urine.
-> So; it can be used in fluid overload condition.
-> ex. after blood transfusion.

ADR :-

1.] Hypokelemia [↓ K^+ Concentration in blood]

-> Furosemide increase K^+ excretion.

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ans 2.) Hyponatremia [↓ Na^+ Concentration in blood] :-

-> Furosemide increase Na^+ excreation.

ans 3.) Hypocalcemia ; Hypomagnesemia :-

-> Because of increased excreation of Ca^{+2} and Mg^{+2}.

ans 4.) Hyperglycemia :-

-> Because of decrease insuline.

ans 5.) Hyperlipidemia :-

-> Because of increase LDL concentration.

ans 6.) Hyperuricemia :-

-> Because of decrease uric acid excreation
-> Contreindicated in patient of gout.

ans 7.) OTO toxicity :-

-> Because of ion imbalance.
-> Tinnitus (Ringing of ear).
-> Vertigo

ans 8.) Cardiac arrythmia :-
-> Because of hypokalemia.

ans 9.) Hypersensitivity Reaction :-
-> Because of Rashes and Itching.

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Que:- Thiazide Diuretics ?

ans M.O.A. :-

→ Thiazides inhibit $Na^+ - Cl^-$ symport channel at DCT of nephrone.
→ So, absorption of Na^+, Cl^- reduced from nephrone.
→ So, water is excreated along with ions.
→ Volume of urine is increased.
→ That is known as **Diuresis.**
→ It is having less efficacy than furosemide.
→ Na^+ is exchanged by K^+.
→ So, K^+ is excreated more which recusults into hypokelemia.
→ Other ions HCO_3^+, Mg^{+2} also excreated in urine.

ans Uses:-

⇒ 1.] Hypertension:-

→ Thiazides increase excreation of Na^+ ; Cl^-.
→ Fluid is more exereated in urine.
→ So, intravascular fluid is reduced.
→ So, Cardiac output is reduced.
→ and blood pressure is also reduced.
→ Thiazides used as long term therapy for hypertension.
→ They are also combined with other anti-hypertensive drugs.

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ans 2.) Oedema:-

→ Thiazides increase urine volume excretion.
→ Intravascular fluid reduced.
→ Fluid shift from ECF to intravascular compartment.
→ And oedema is reduced.
→ Used in cardiac Oedema ; Hepatic oedema ; Pulmonary Oedema.
→ Thiazides are low efficacious than furoside in reducing oedema.

ans 3.) Hypercalcaemia:-

→ When calcium is excreated more into the urine known as hypercalcemia.
→ Thiazides reduce calcium excreation.
→ So, it can be used in hypercalcemia.

ans Diabetes Insipidus:-

→ Thiazide reduce frequency and water filtration in urine in case of Diabetes insipidus.
→ Mechanism of this effect is not properly understood.

~~ans Diabetes Insipidus:-~~
→ Thiazide reduce frequency and water filtration in urine in case of Diabetes insipidus.
→ Mechanism

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ans Probable Explanation :-

→ Thiazides increase excreation of Na^+ ; Cl^- along with water.
→ Electrolytes in blood reduced.
→ They are more absorbed from proximal tubule.
→ water along with that ion also more absorbed.
→ Diabetes insipidus symptoms reduced.

ans ADR:-
→ i.) Hypokelemia :- [↓ K^+ Conc. in blood]
- Thiazide increase K^+ excretion.

→ ii.) Hyponetremia :- [↓ Na^+ Conc. in blood]
- Thiazide increase Na^+ excreation

→ iii.) Hypomagnesemia :-
- Bcz of increase excretion of Ca^{+2} and Mg^{2+}

→ iv.) Hyperglycemia :-
- Bcz of decrease insuline.

→ v.) Hyperlipidemia :-
- Bcz of increase LDL concentration.

→ vi.) Hyperuricemia :-
- Bcz of decrease uric acid excreation.
- Contraindicated in patient of gout.

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Que:- Aldosterone Antagonist ?
or
K^+ sparing diuretics ?

=> Name of the drugs:-
- Spironolectone.

=> M.O.A.:-

~> Aldosterone Action:-

-> Aldosterone is a mineralocorticoids.
-> It reach into the cells of DCT.
-> It binds with it's intracellular receptors.
-> Stimulates DNA ; transcription occurs and RNA formed.
-> Translation occurs and protein formed.
-> This protein absorbed Na^+ and excreate K^+ into urine.

~> Spironolectone Action:-

-> Spironolectone binds with aldosterone receptors.
-> Blocks the action of aldosterone.
-> So, Na^+ is excreated more into urine.
-> Water absorption from urine reduced.
-> Urine volume increased.
-> K^+ excreation is reduced
-> So, it is K^+ sparing diuretics.

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=> Uses:-

-> It is used along with other diuretics (Thiazide / furosemide)
-> Thiazides / furosemide increase K^+ excretion and hypokelemia occurs.
-> Spironolectone reduce K^+ excretion.
-> So, hypokelemia is reduced.

=> 2) Oedema:-

=> 3) Hypertension:-
-> Resistant hypertension. (In case of hyperaldosteronism

=> ADR:-
- Nausea
- Vomiting
- Diarrhea
- Gynecomastia
- Loss of Libido.

SQ:- Spironolectone should not be combined with ACE inhibitor / ARB ?

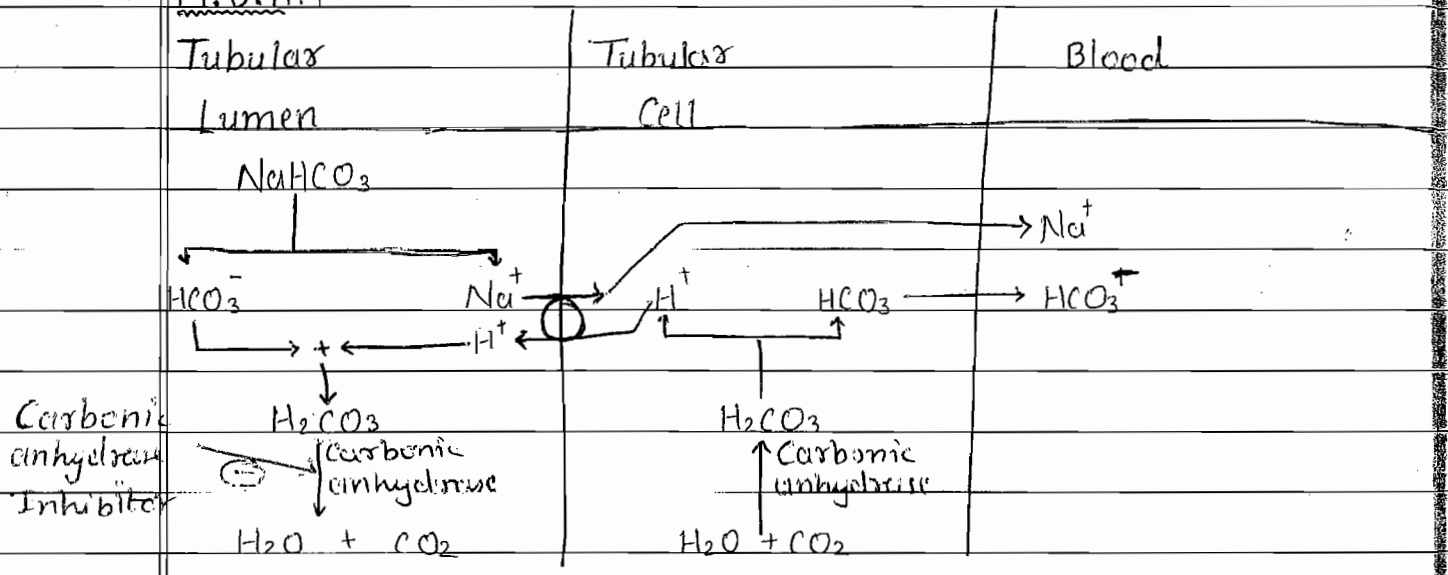
-> Because ACE inhibitor / ARB produce hyperkelemia
-> Spironolectone also produce hyperkelemia.
-> So; combination of both leads to Cardiac arrythmia.

Que:- Carbonic anhydrase inhibitor (Antagonist) ?

ans Name of Drugs :-

- Acetazolamide
- Dorzolamide

ans M.O.A. :-



→ In tubular cells of PCT ;
H_2O and CO_2 combines with the help of carbonic anhydrase enzyme and form H_2CO_3.

→ H_2CO_3 dissociated into H^+ and HCO_3^-.

→ H^+ is exchanged with Na^+. via

$Na^+ - H^+$ antitransport (antiport) channel.

→ Remaining HCO_3^- of tubular lumen binds with H^+ and form H_2CO_3.

→ This H_2CO_3 is dissociated into H_2O and CO_2 with the help of carbonic anhydrase.

→ H_2O and CO_2 diffuse into Tubullar cells.

→ Na^+ and HCO_3^- of Tubullar cells deffuce into blood.

→ along with them water is also absorbed.
→ Carbonic anhydrase inhibitor inhibit carbonic anhydrase enzyme and prevent the binding of H_2O and CO_2
→ So ; Na^+ and HCO_3^- ions reabsorption reduced.
→ They pass into urine.
→ So, Diuresis occurs.

aw Uses :-

aw i.) As Diuretic :-
→ CA inhibitor is weak diuretic.
→ It is used as diuretic along with other diuretics.

aw ii.) Glaucoma :-
→ CA inhibitor reduce aquous humor production.
→ So, reduce intraocular tension.
→ So, used in glaucoma.

aw iii.) Alkalisation in Urine :-
→ CA inhibitor decrease $NaHCO_3$ absorption from PCT.
→ So ; increase excretion of Na^+ and HCO_3^-.
→ because of that urine becomes alkali.
→ So ; it is used in acidic drug poisioning.

aw iv.) Mountain Sickness :-
→ For symptomatic relief by reducing CSF production.

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ans ADR:-
-> Drowsiness
-> Metabolic acidosis
-> Hypokalemia
-> Headache
-> Renal stone.

Que:- Osmotic Diuretics ?

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ans Name of Drugs:-
- Mannitol
- Glycerol

ans M.O.A.:-
-> Osmotic diuretics are present in intravascular compartment.
-> They create osmotic pressure.
-> Increase water absorption from intestitial space to intravascular compartment.
-> They are freely filtered into urine.
-> Increase the volume of urine.
-> That's how diuresis occurs.
-> They decrease oedema.

ans Uses:-

ans i.) Oedema:-
-> Mannitol is preferred in Oedema.
-> It is preferred in cerebral oedema after head injury.
-> because it prevents rise in intracranial tension.

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ans ii.) Glaucoma:-

-> In acute angle closer glaucoma;
IV mannitol is used to reduce intra ocular tension. (pressure).

ans iii.) Maintainance of volume of intravascular fluid :-

-> after dialysis

-> During cardiovascular shock ; prevent Renal damage.

ans ADR:-

- Headache
- Nausea ; Vomiting
- Pulmonary Oedema
  - Due to rapid use.

★Central Nervous System★

Quer- **Anticliuretics :-**

→ These are the substances which reduce urine volume.

:SN:- **Vasopressin OR ADH :-**

→ It is produce in supraoptic nucleus of hypothalamus.
→ Stored into posterior pituitary gland.
→ It is peptide hormone.

ans **M.O.A. :-**

→ **V_1 receptor mediated :-**

Vessels -> Vasoconstriction
↓
BP ↑

GIT → Peristalsis ↑
Platelets → Aggregation
Uterus → Contraction

→ **V_2 mediated action :-**

Kidney → Vasopressin
reach at collecting tubule
→ Increase production of water in channel.
→ Increase absorption of water from urine into blood.

→ **Regulation in Body :-**

Body osmolarity ↑ → ADH ↑
Body osmolarity ↓ → ADH ↓

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Q) Desmopressin :-

→ It's a synthetic analogue.

→ It is V_2 receptor agonist.

→ It is more potent than vasopressin.

→ It can be used by oral ; nasal and parentral route.

Q) Terlipresin :-

→ Prodrug in vasopressin.

→ It is having selective V_1 agonist action.

Q) Uses :-

Q) i.) Diabetes Insipidus :-

→ It occurs because of deficiency of ADH.

→ So ; water content in urine is increase.

Polyuria :-

→ Increased frequency and volume of urine.

→ Only water is excreated.

→ It is two types.

1.) Pitutary DI (Central) :- ADH concentration is reduced.

2.) Nephronic DI (Renal) :- ADH is normal but Renal tubules are resistance to ADH. (not responding).

→ Desmorphine is used in the treatment of DI. It will reduce urine volume by absorption of water from urine to blood.

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ans ii) Nocturnal enuresis:-

-> Desmopressin can be used in nocturnal enuresis.

-> It reduce urine volume and decrease the chances of passing urine in blood.

ans iii) Oesophageal Variesis:-

-> Bleeding from oesophagus due to rupture of vessels.

-> Which mostly occurs in portal hypertension

-> Blood vomiting occurs.

-> Terlipresin can be used to produce vasoconstriction of vessels and reduce bleeding.

ans ADR:-

-> Nausea

-> Vomitting

-> Hypertension

-> Increase chances of angina.

-> Diarrhea

-> Increase thrombosis.

Que:- General Anasthesia:-

ans Features of GA:-

- Reversible loss of conciousness.
- Reversible loss of sensations.
- Analgesia.
- Amnesia.
- Muscle Relaxation.
- Loss of reflexes.

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as Classification of GA:-

as Inhalation:-

→ Gas:-
- Nitrous Oxide

→ Volatile Liquid:-
- Halothane
- Isoflurane
- Desflurane
- Sevoflurane.
- Ether

as Intravenously:-

→ Inducing agent:-
- Thiopentone Sodium
- Propofol
- Etomidate.

as Slowly acting:-

1.) Benzodiazepines:-
- Diazepam
- Lorazepam
- Midazolam

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2.) Dissociate anaesthetic:-
- Ketamine.

3.) Opioids:-
- Fentanyl
- Alfentanyl
- Remifentanyl
- Sufentanyl.

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Ques- Stages of Anesthesia?

→ There are four stages of anesthesia.

ans Stage - I :-

→ Stage of analgesia.

→ Person is concious but drowsy.

ans Stage - II :-

→ Stage of ~~excitement~~ excietment.

- Patient loss conciousness.
- Sympathetic activity increased.
  - HR ↑
  - BP ↑
  - Pupil → dilatation.
  - Muscle tone → increase
  - Breathing → irregular.

ans Stage - III :-

→ Stage of surgical anesthesia.

- Reflexes lost.
- breathing regular
- Intercostal muscle → paralysed.

ans Stage - IV :-

→ Stage of medullary paralysis.

- Respiratory centre → depressed.
- Vasomotor centre → depressed.
- Person may go into the stage of coma.
- Death can also occurs.

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Quer- Ether :-

→ Volatile liquid.

ans Characteristic :-

bad
- Irritant
- Explosive
- Inflammable

Good
- Very good analgesic
- Potent anesthetic
- It is having wide margin of ~~safety~~ safety.
- It is cheep.
- It does not sensitise the heart for catecolamine. (adrenaline, noradrenaline).

ans ADR :-
- Post operative nausea and vomiting.
- Less hepatotoxicity.

Quer- Halothane :-

→ It is volatile liquide.

ans M.O.A. :-
→ It is faster absorbed from alveoli.
→ Mix into blood and reach CNS.
→ It reduce the neuronal activity.
→ and produce anesthesia.

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ans Benefits :-

-> Non - irritant.
-> Non - explosive.
-> Well tolerated.
-> Induction & recovery is faster.
-> Prefered in children.

ans Drawbacks :-

-> Poor analgesic.
-> It sensitised the heart for catecolemines and produce cardiac arrythmia.
-> Respiratory depression.
-> Hepatotoxicity.
-> It's costly.

Que :- Nitrous Oxide (N_2O) ?

-> It is a gasseous general anesthetic.

ans M.O.A. :-

-> It is absorbed faster from alveoli.
-> absorbed into blood.
-> It reduce neuronal activity. (Block impulses).

ans Benifits :-

-> Induction and recovery is faster.
-> Non - irritant.
-> Non - inflammable.
-> Wide margin safty.
-> Excellent analgesic.
-> Having less effect on heart and blood pressure.
-> It's cheap.

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Ans Disadvantages:-

-> Poor anesthetics.
-> Poor muscle relaxent.
-> It produce diffusion hypoxia.
-> increase intracranial tension.

Quer- Hallothane is combined with nitrous oxide?

-> Dose requirement of Hallothane for production of anesthesia is reduced.
-> Hallothane is potent anasthetic but poor analgesic.
-> Nitrous oxide (N_2O) is potent analgesic but poor anasthetic.
-> Combination is beneficial as effects are combined.
-> Nitrous oxide (N_2O) is rapidly diffuse and it also increase diffution of halothane. earlier effects are produced.

Quer- Propofol:-

-> It is a parentrally (intreivenously) used general anesthetic.

Ans M.O.A.:-

-> It binds with GABA channel.
-> Increase Cl^- current (Conc.) is neurons.
-> Hyperpolerisation of neurons.
-> Decreass neuronal activity. (↓ impulse generation).

Ans Benefits :-

-> Induction and recovery is faster.
-> Non - irritant.
-> Non - inflammable
-> Good for outpatient surgery (day care surgery).
-> Good for producing sedation in ICU.
-> Can be used for controlling convulsion in case of status epilepticus.
-> Preffered in asthmatics.

Ans Disadvantages :-

-> Fall in blood pressure.
-> Pain at injection site.
-> Acidosis.
-> Respiratory depression.

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Que:- Thiopentone Sodium ?

-> Ultra short acting barbiturates.

Ans M.O.A :-

-> It acts on barbiturates receptors of GABA channel.
-> Increase chloride current.
-> Neuronal potential becomes negetive.
-> Impulse generation reduced.

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Benefits / Advantages / Uses :-

-> It is rapidly acting.
-> Used as induction of anesthesia.
-> It also work as anti-convulsant.
-> at low dose it can be used as Narcoanalysis.
-> It's action is terminated fast.

ADR / Disadvantages :-

-> Respiratory depression.
-> Depression of vasomotor centre.
-> Poor analgesic
-> Poor muscle relaxent.

Redistribution of Thiopentone Sodium ?

-> First thiopentones reaches to the organs having higher blood flow. (brain)
-> Then, it is redistributed to the organs like muscle ; fat.
-> So ; it's action is terminated.

Que :- Ketamine ?

-> It is "dissociated anesthesia".
-> It is good analgesic.
-> It produce sedation and amnesia.

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ans M.O.A :-

-> Ketamin acts on NMDA (N-methyl D-aspartase) type of glutamate receptor.
-> Block the action of glutamate.
-> Reduce neuronal activity.

ans Advantages :-

-> Bronchodilatation -> Preffered in asthma patient.
-> Do not produce hypotension
- So, can be used in hypovolumic patients.
-> It is well tolerated by children.
-> Preffered for surgery of head & Neck region.
-> It can be used by $\vec{IV}$; $\vec{IM}$; oral or rectal route.

ans Disadvantages :-

-> Hypertension , increase HR
-> Hallucination
-> delirium (false beleive)
-> increase intracranial tension.

Que: Preanesthetic Medication :-

-> The drugs which are given before anesthesia to make anesthesia safe and plesent known as preanesthetic medication.

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aw For reduction of secretion (Antisecretory):-

-> Glycopyrolate which is anticholinergic used to reduce secretions.
-> Anesthetic drugs increase nasopharyngeal secretions.
-> and increase the chances of aspiration.
-> So, antisecreatory drugs are used as preanesthetics medications.

aw For reducing anxiety and reducing sleep:-

-> Benzodiazepams (Diazepam ; Lorazepam) can be used to reduce anxiety and increasing sleep in patient.

aw For reducing pain preoperatively as well as postoperativly:-

-> Opioids (fentanyl ; Ramifentanyl) can be used to reduce pain.
-> NSAIDs (Diclofenac) can also be used for low grade of pain.

aw For prevention of Nausia and vomiting:-

-> Antimitis Drugs - 5HT3 antagonist ondensetron is used.
-> It is vary effective and safe for prevention of Nausea and vomiting.

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ans For increasing gastric emptying :-

-> Prokinetic drug -> Metoclopramide is used.
-> It increase the gastric emptying.

ans For reducing acid secretion :-

-> Proton pump inhibitors -> Omeprazol ; Rabeprasol ; Or pentoprazol can be used.
-> They reduce acid production.
-> Reduce the chances of aspiration pneumonia.

Que:- Classification of Local anesthetics ?

-> According to clinicle use.

ans i.) Surface anesthesia :-
- Cocaine
- Lignocaine
- Tetracaine
- Benzocaine.

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ans ii.) Injectable anesthesia :-
-> a.) Short acting with low potency :-
- Procaine
- Chloroprocaine
-> b.) Intermediate acting with intermediate potency :-
- Lignocaine
- Mepivacaine.
-> c.) Long acting with high potency :-
- Tetracaine
- Ropivacaine.
- Bupivacaine